UG10092

MCX N Series Hardware Design Guide

Rev. 6.1 — 30 June 2026 **User guide**

Document information

Information	Content
Keywords	UG10092, MCX N Series of MCUs, (MCX N, N94, N54, N53, N52, N24, N23), 150 MHz, dual-core, high-performance, secure, low-power, general purpose MCUs, Arm Cortex-M33 MCU, advanced security, eIQ Neutron Neural Processing Unit (NPU), EdgeLock Secure Enclave, Artificial Intelligence (AI), Machine Learning (ML)
Abstract	This document aims to help hardware engineers design and test their MCX N series MCU-based designs. It provides an overview of the design features and includes design recommendations and checklists.

1 Introduction

This document aims to help hardware engineers design and test their MCX N series MCU-based designs. It provides information about board layout recommendations and design checklists to ensure first-pass success and avoid board bring-up problems. For the relevant device-specific hardware documentation, see MCX N Series Microcontrollers.

This document includes the following sections:

1. Section 1 "Introduction"
2. Section 2 "Architecture overview"
3. Section 3 "Recommended schematic, layout, and special considerations"
   - VFBGA184 package
   - HLQFP100 package
   - HDQFP172 package
4. Section 4 "Bulk and decoupling capacitance"
5. Section 5 "Power configurations"
6. Section 6 "DCDC"
7. Section 7 "Clocks"
8. Section 8 "Digital interfaces"
9. Section 9 "Analog interfaces"
10. Section 10 "HMI interfaces"
11. Section 11 "Unused pins"
12. Section 12 "Design checklists"
13. Section 13 "References"
14. Section 14 "Acronyms and abbreviations "

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2 Architecture overview

Before designing a system, one must understand the power architecture and how the core domain is powered up. To perform the schematic process correctly, understanding the power architecture is essential. Figure 1 shows the chip level architecture and not the package pinout. Packaging differences affect the schematic design and these differences are discussed later in this document.

Note: *Die pins are shown in the figure below. Ensure to check your package-specific pinout, as in some packages, these die pins are double bonded to package pins.*

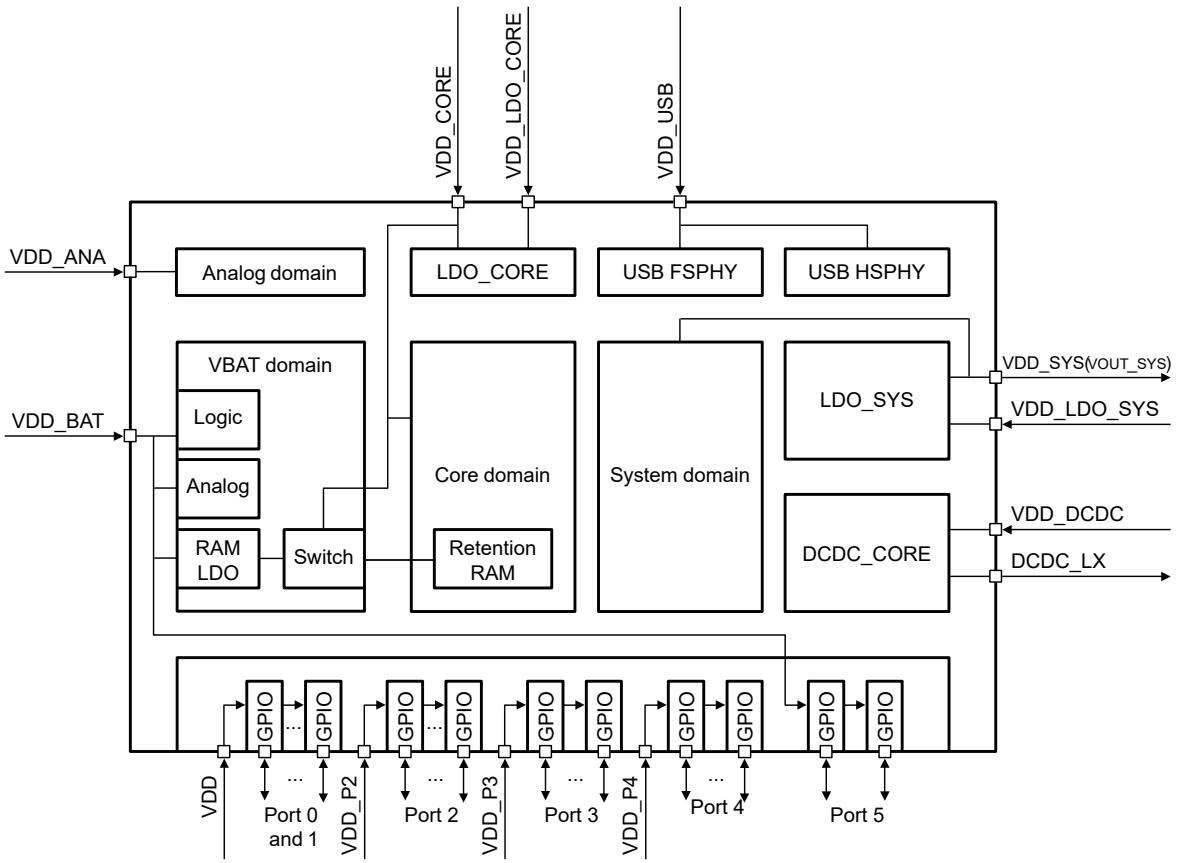


Figure 1. MCX N series power architecture

MCX N series of devices are designed to allow for flexibility in the power schema for the device. There are eight independent power domains to consider. Each power domain has its own operating ranges and restrictions, see Table 1.

Table 1. Power domain operating requirements

Power domain	VFBGA184 and HDQFP172	HLQFP100	Voltage range	Remarks
Core	LDO configuration: VDD_LDO_CORE DCDC configuration: VDD_DCDC	LDO configuration: VDD DCDC configuration:	1.71 V to 3.60 V	This range applies to both DCDC and core LDO regulator inputs.

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Table 1. Power domain operating requirements*...continued*

Power domain	VFBGA184 and HDQFP172	HLQFP100	Voltage range	Remarks
		VDD_DCDC		
System	VDD_LDO_SYS	VDD_DCDC	1.86 V to 3.6 V	This pin is internally bonded to the VDD_DCDC pin in HLQFP100100. As such, HLQFP100100 designs **MUST ALWAYS** power VDD_DCDC.
	VDD_SYS	VDD_SYS	• 1.71 V to 1.98 V • 2.25 V to 2.75 V (Fuse programming only)	This pin is internally tied to the output of the LDO_SYS regulator. It must be connected to a 2.2 µF capacitor to the ground. Tying the VDD_SYS pin to the VDD_LDO_SYS pin bypasses this regulator and is only allowed in both BGA and HDQFP172 packages. **However, it is not recommended to bypass this regulator.** ***Important:*** *If the internal LDO_SYS regulator is bypassed, user software and power supplies must conform to the restrictions on this range that are specified in the data sheet.* ***Important:*** *The extended voltage range of 2.25 V to 2.75 V is for fuse programming only. User software must ensure that the maximum allowable time at this voltage (specified in the data sheet) is not exceeded.*
VDD	VDD	VDD	1.71 V to 3.60 V	It also powers VDD_P2 in the HLQFP100 package.
VDD_P2	VDD_P2	VDD	• 1.71 V to 3.60 V • 1.14 V to 1.32 V	The extended voltage range of 1.14 V – 1.32 V is allowed, provided this rail ramps after the system domain. The extended voltage range of 1.14 V to 1.32 V is not allowed in the HLQFP100 package because it is powered by the VDD pin.
VDD_P3	VDD_P3		• 1.71 V to 3.60 V • 1.14 V to 1.32 V	The extended voltage range is allowed, provided this rail ramps after the system domain.
VDD_P4/ VDD_ANA	VDD_P4 and VDD_ANA	VDD_P4 and VDD_ANA	1.71 V to 3.60 V	VDD_P4 and VDD_ANA must be tied together through an (optional) filtering inductor.
VDD_BAT	VDD_BAT	VDD_BAT	1.71 V to 3.60 V	This rail is required and must ramp before or with the system domain.
USB	VDD_USB	VDD_USB	3.00 V to 3.60 V	This rail is optional if USB is not used in the application. If USB is not used, ground this rail through a 10 kΩ resistor.
ADC reference	VREFH	VREFH	0.99 V to VDD_ANA	This pin must be connected to VDD_ANA if not used.

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Table 1. Power domain operating requirements*...continued*

Power domain	VFBGA184 and HDQFP172	HLQFP100	Voltage range	Remarks
	VREFL	VREFL	(VSS_P4 - 0.1 V) to (VSS_P4 + 0.1 V)	This pin must be connected to the same potential as VSS_P4 even if the ADC is not used.

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3 Recommended schematic, layout, and special considerations

The MCX N series device is available in three packages listed below. To kick-start your design, click the hyperlink that corresponds to your device package:

- VFBGA184
- HLQFP100
- HDQFP172

This section discusses the package dimensions, specifications, recommended general layout, recommended route-out, and PCB stack-up.

3.1 VFBGA184 package

This section provides details about the VFBGA184 package outline, placement, routing, and PCB stack-up.

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3.1.1 Schematic

This section provides details about the recommended schematic options. There are two basic schematic drawings that can be used for the VFBGA184: LDO configuration or DCDC configuration. In the LDO configuration, the VDD_CORE domain is powered by the Core LDO regulator. In the DCDC configuration, the VDD_CORE domain is powered by the DCDC regulator. These two schematic drawings are presented below.

3.1.1.1 LDO configuration: minimal component and minimal size design

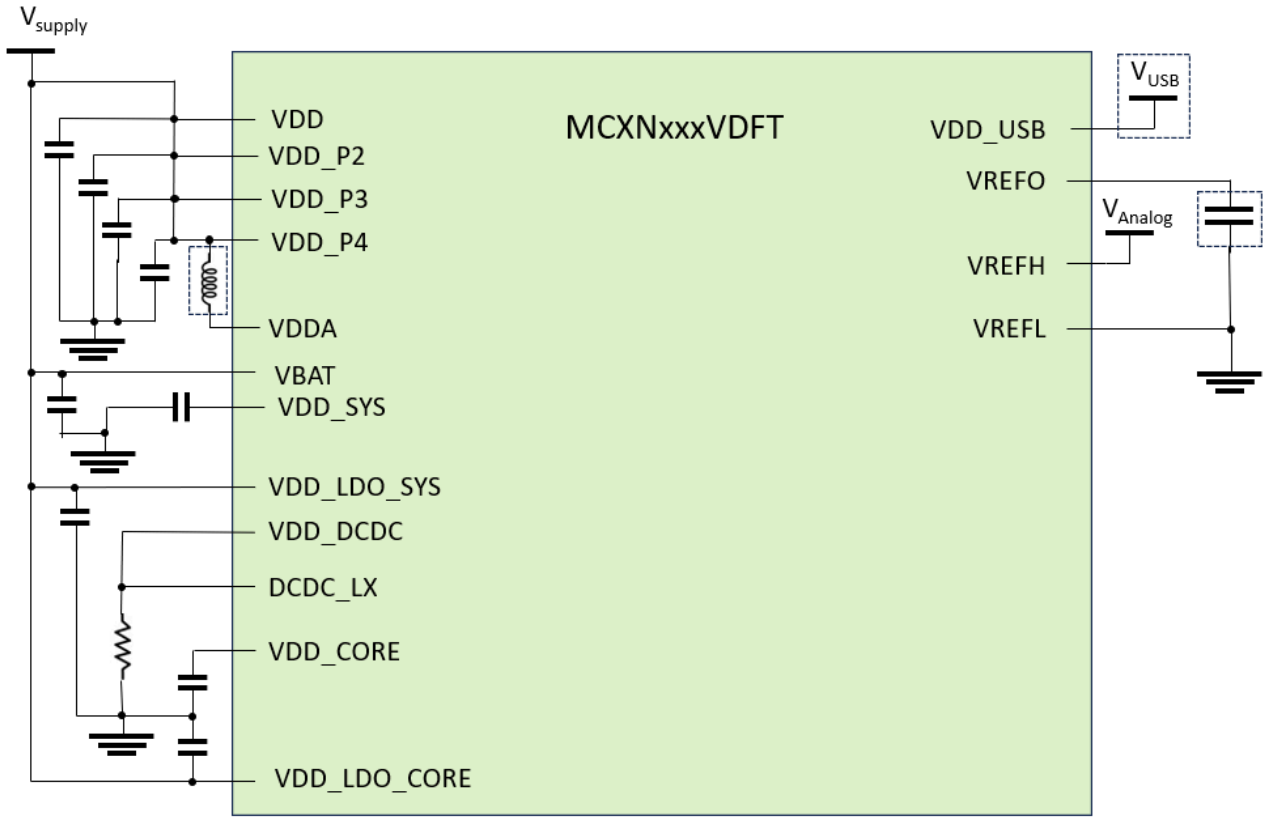


Figure 2. LDO VFBGA configuration

Note: *For more information on bypass and bulk capacitance number and values, refer to the section Bulk and decoupling capacitance in this document.*

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3.1.1.2 DCDC configuration: Power efficient design

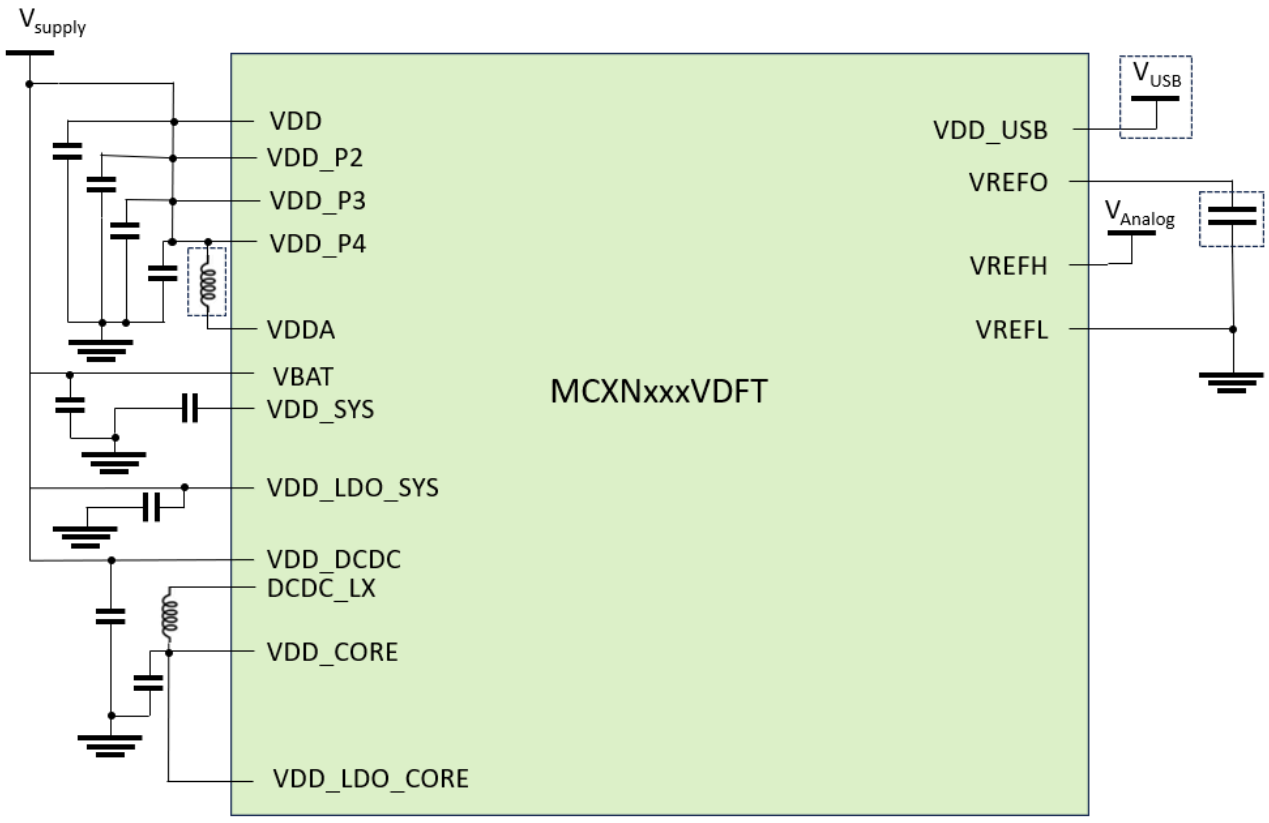


Figure 3. DCDC VFBGA configuration

Note: *For more information on bypass and bulk capacitance number and values, refer to the section Bulk and decoupling capacitance in this document.*

3.1.2 VFBGA184 layout

As the placement of components is the first step in the layout process, it is important to discuss key peripheral placement. The key peripheral placement of the VFBGA184 package is shown in Figure 4. This placement allows for the most efficient routing of key peripherals in your design.

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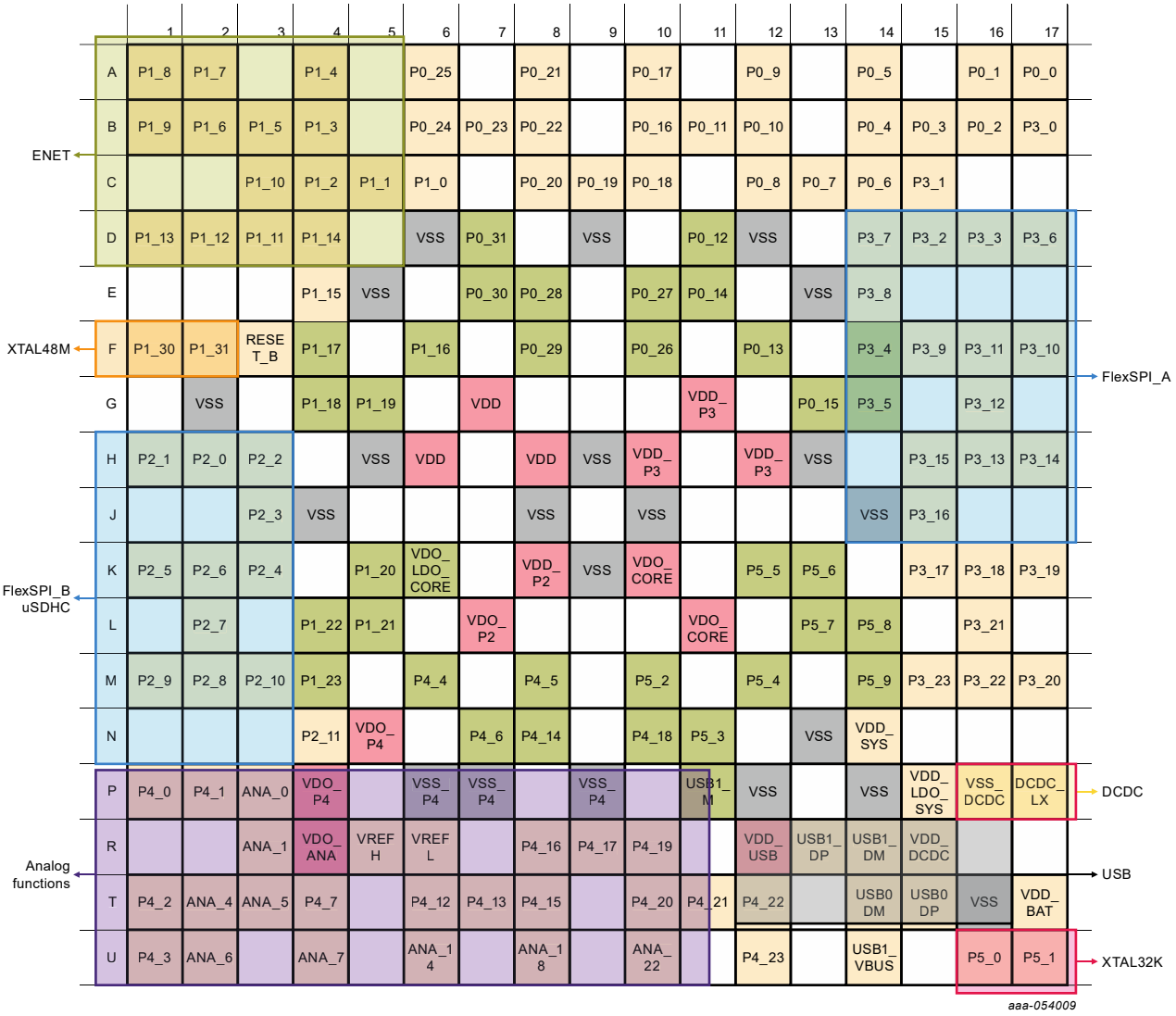


Figure 4. Key peripheral placement for VFBGA

After placement is set, the routing process starts. The MCX N series VFBGA184 is designed to allow for routing all 184-pins externally in a four-layer design. The design uses 4 mil trace widths and standard PCB technology (no blind vias or micro vias are necessary). The recommended stack-up is shown in Figure 5.

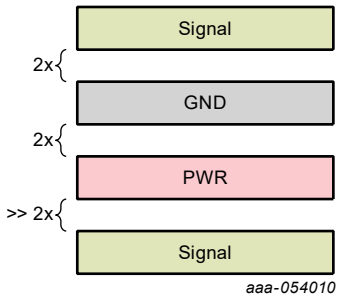


Figure 5. Recommended four-layer stack-up

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3.1.3 VFBGA184 routing

The recommended route-out is shown in Figure 6.

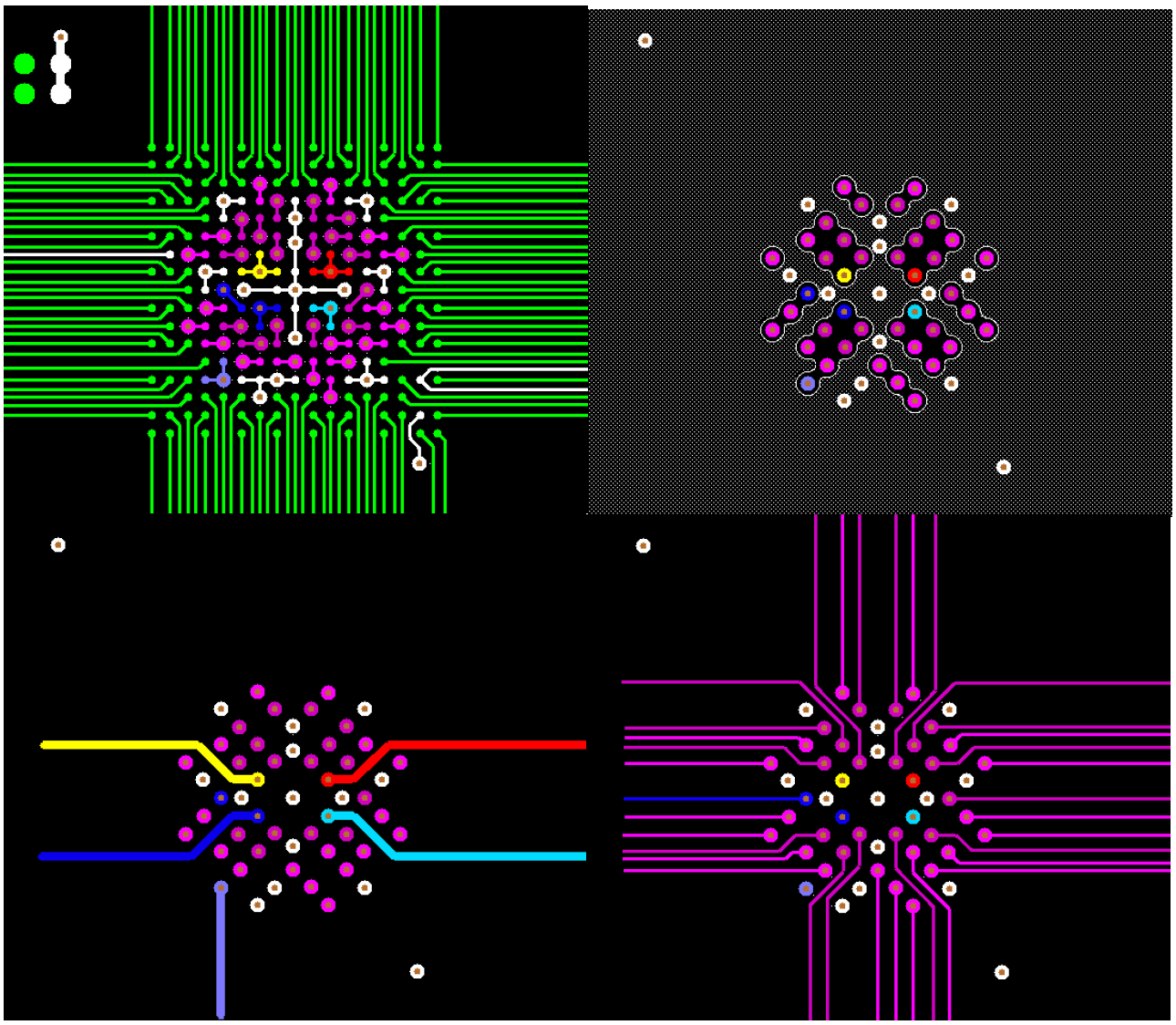


Figure 6. Recommended VFBGA escape routing

3.1.4 Package outline

The VFBGA184 package is documented in the SOT2172-1 package drawing found online at www.nxp.com. The SOT2172-1 document contains all the necessary information to create a complete footprint for your design and provides solder mask template information.

3.1.5 Bulk and decoupling capacitor placement

Like other high-speed BGA devices, decoupling and bulk capacitors must be placed on the bottom side of the PCB, directly underneath the device. Using the smallest package type allowed by budget and manufacturing constraints for both capacitor types is recommended. These capacitors must be mounted as close to the

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power vias as possible. It minimizes the inductance and ensures high-speed transient current demand by the processor.

An example layout is shown in Figure 7.

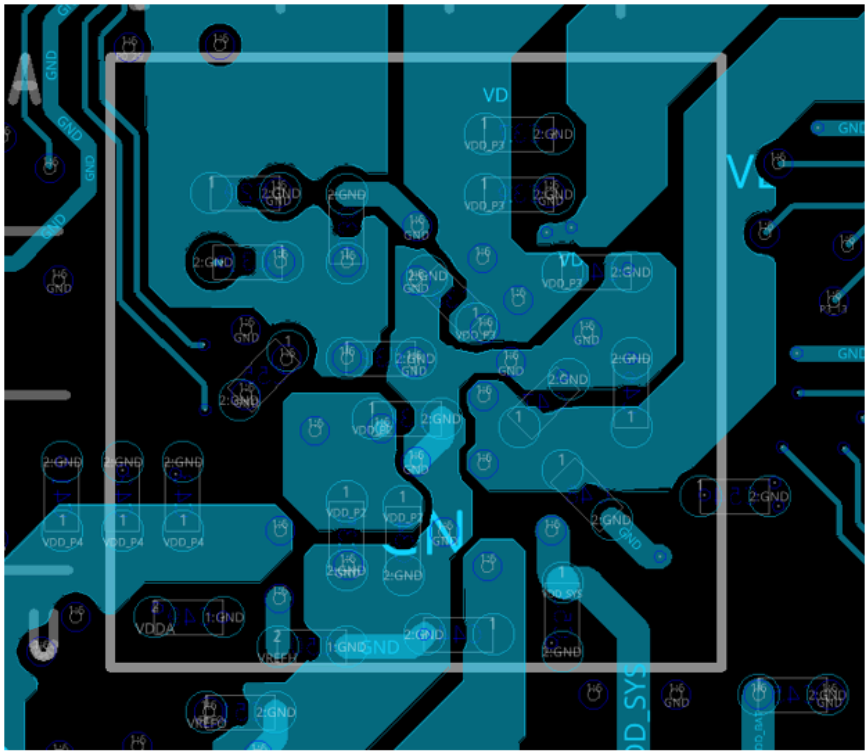


Figure 7. Example decoupling and bulk capacitor layout for VFBGA package

3.2 HLQFP100 package

This section provides details about the HLQFP100 package outline, schematic, placement, routing, and PCB stack-up.

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3.2.1 Schematic

This section provides details about the recommended schematic options. There are two basic schematic drawings that can be used for the HLQFP100: LDO configuration or DCDC configuration. In the LDO configuration, the VDD_CORE domain is powered by the Core LDO regulator. In the DCDC configuration, the VDD_CORE domain is powered by the DCDC regulator. These two drawings are presented below.

3.2.1.1 LDO configuration: minimal component and minimal size design

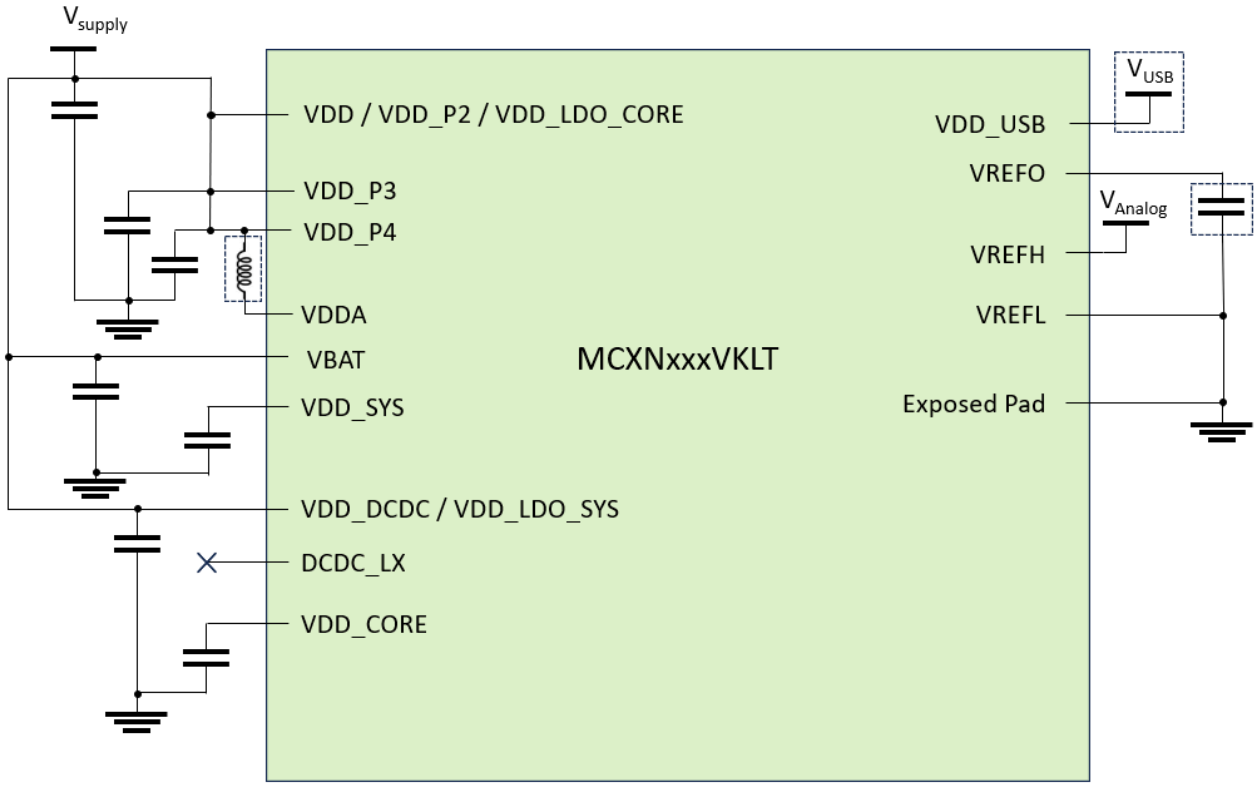


Figure 8. HLQFP100 LDO configuration

Attention: *For more information on bypass and bulk capacitance number and values, refer to the section Bulk and decoupling capacitance in this document.*

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3.2.1.2 DCDC configuration: Power efficient design

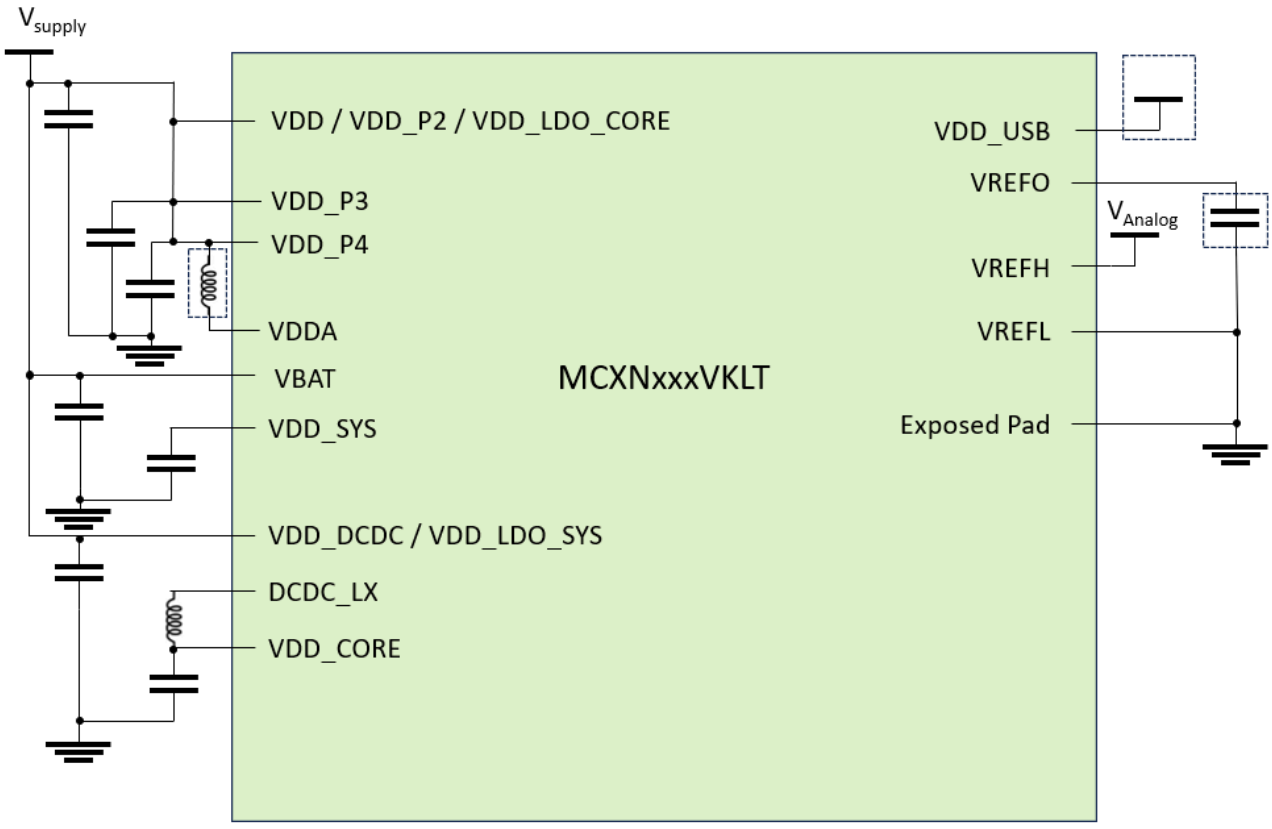


Figure 9. HLQFP100 DCDC configuration

Attention: *For more information on bypass and bulk capacitance number and values, refer to the section Bulk and decoupling capacitance in this document.*

3.2.2 Package outline

The HLQFP100 package is documented in the SOT2229-1 package drawing found online at www.nxp.com. The SOT2229-1 document contains all the necessary information to create a complete footprint for your design and provides solder mask template information.

3.2.3 HLQFP placement

The key peripheral placement of the 100-pin HLQFP package is shown Figure 10.

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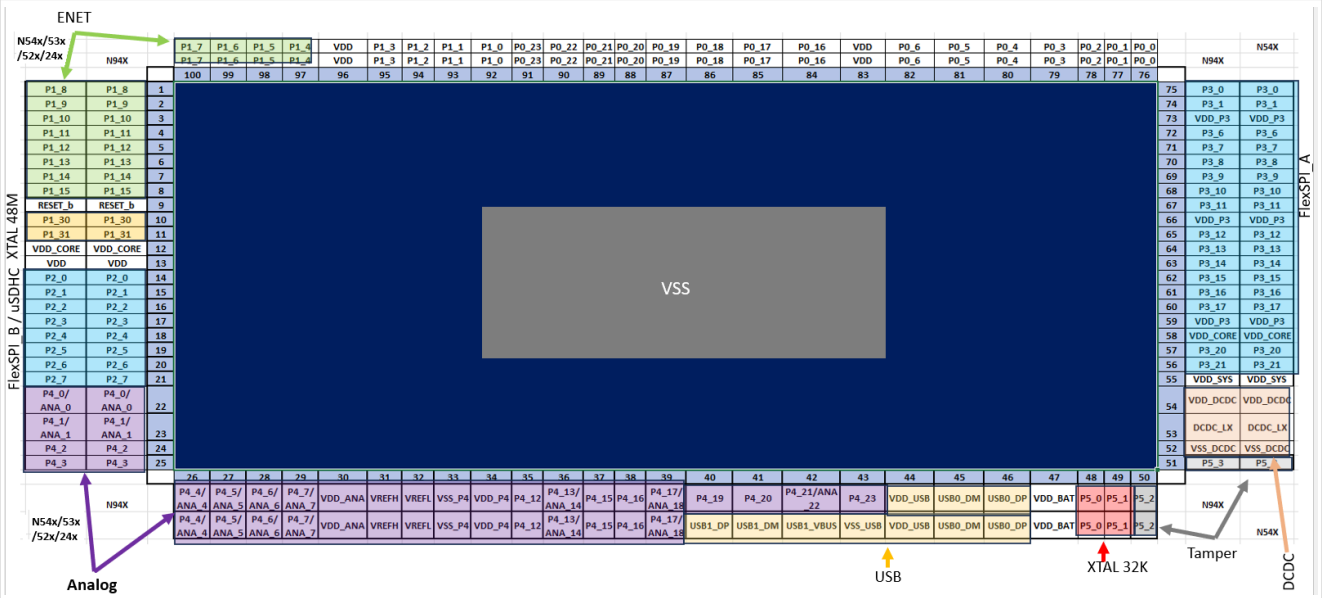


Figure 10. Key peripheral placement of HLQFP package

3.2.4 HLQFP routing

Due to the nature of the HLQFP package, a complicated route-out similar to the VFBGA is not necessary. All traces can be routed to the top layer of the PCB. **However, the HLQFP has an exposed pad on the bottom of the package that also serves as the device ground. This pad must be handled appropriately.** The exposed pad (shown in Figure 11) must be well stitched to the supply ground (or negative supply net). The PCB pad dimensions, solder mask opening pattern, and solder paste stencil are provided in SOT2229-1.

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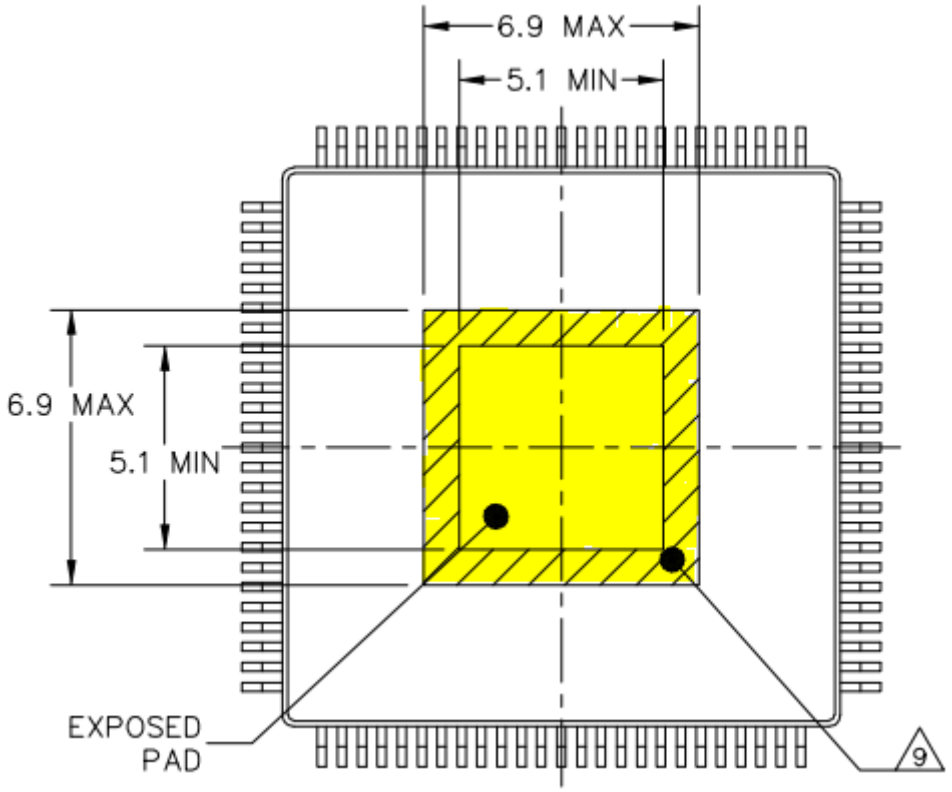


Figure 11. Exposed pad for HLQFP package

Note: *The exposed pad shown in the center of Figure 11 must be grounded for proper operation of the device.*

3.2.5 PCB stack-up for HLQFP100

Due to the nature of HLQFP100 packages, there are no restrictions or requirements on the number of layers used for the HLQFP100 package.

3.3 HDQFP172 package

This section provides details about the HDQFP172 package outline, placement, routing, and PCB stack-up.

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3.3.1 Schematic

This section provides details about the recommended schematic options. There are two basic schematic drawings that can be used for the HDQFP172: LDO configuration or DCDC configuration. In the LDO configuration, the VDD_CORE domain is powered by the Core LDO regulator. In the DCDC configuration, the VDD_CORE domain is powered by the DCDC regulator. These two drawings are presented below.

3.3.1.1 LDO configuration: minimal component and minimal size design

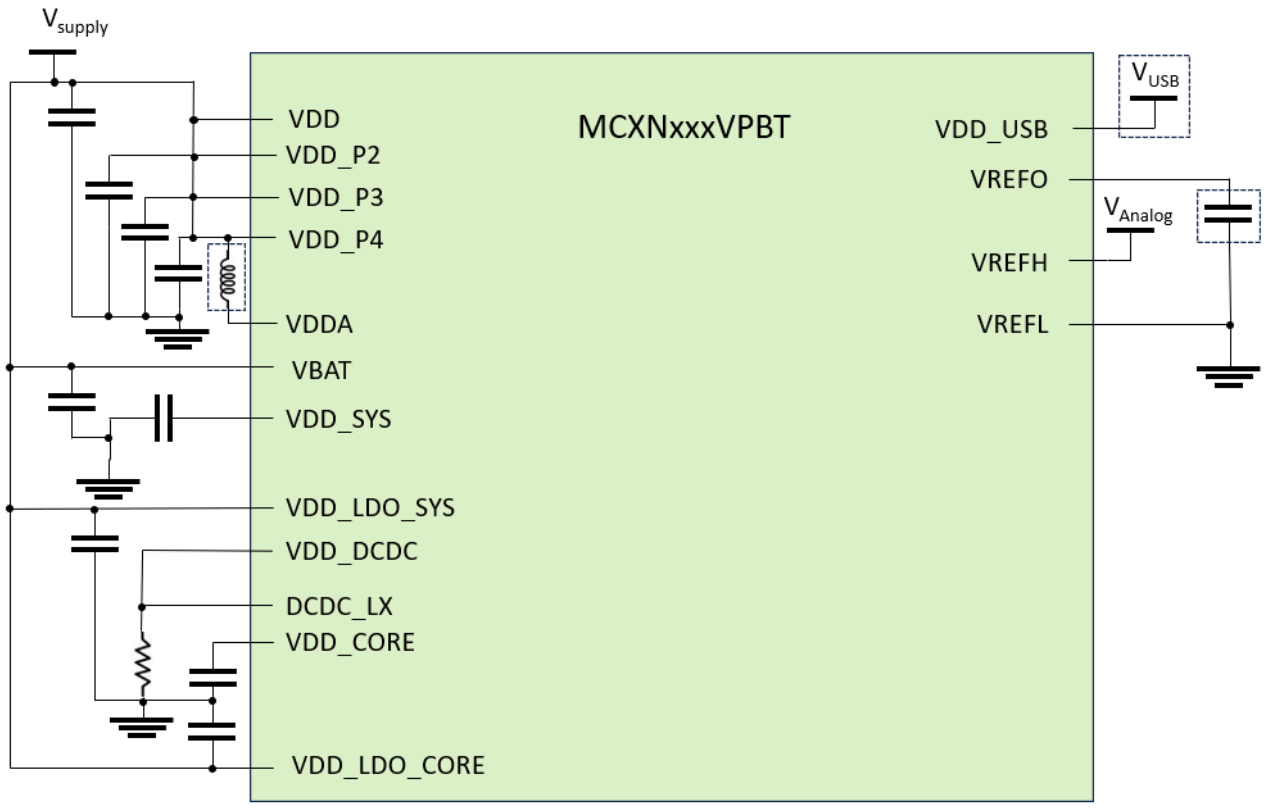


Figure 12. HDQFP172 LDO configuration

Note: *For more information on bypass and bulk capacitance number and values, refer to the section Bulk and decoupling capacitance in this document.*

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3.3.1.2 DCDC configuration: Power efficient design

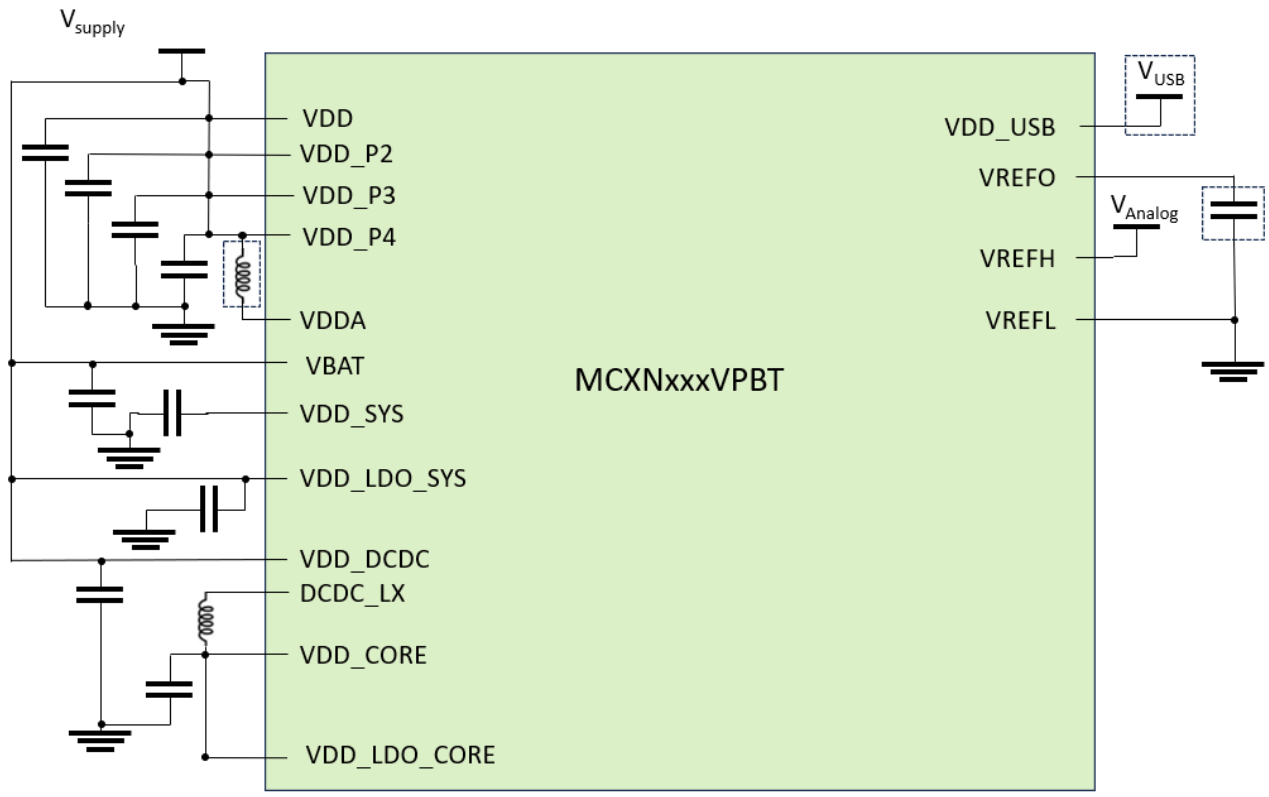


Figure 13. HDQFP172 DCDC configuration

Note: *For more information on bypass and bulk capacitance number and values, refer to the section Bulk and decoupling capacitance in this document.*

3.3.2 Package outline

The HDQFP172 package is documented in the SOT1940-1 package drawing found at www.nxp.com. The SOT1940-1 document contains all the necessary information to create a complete footprint for your design and provides solder mask template information.

3.3.3 HDQFP172 placement

Figure 14 shows the key peripheral placement of the HDQFP172 package.

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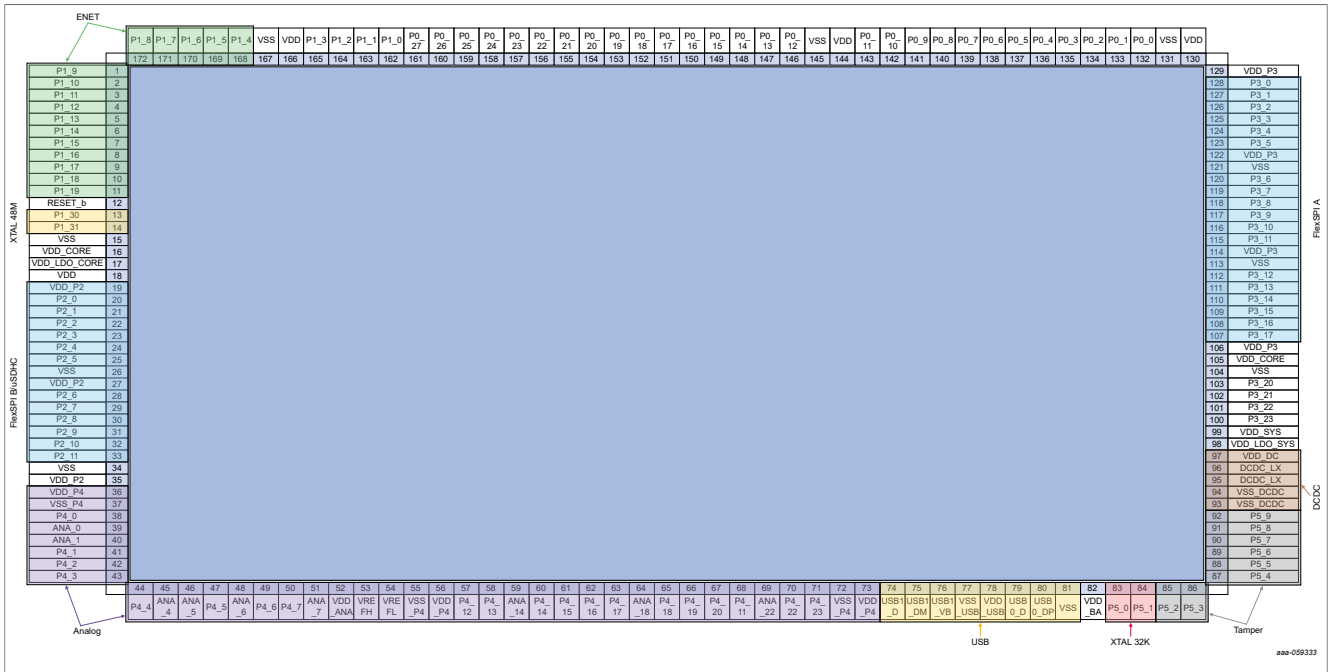


Figure 14. HDQFP172 package placement

3.3.4 HDQFP172 routing

All traces can be routed to the top layer of the PCB. No special routing recommendations are necessary.

3.3.5 PCB stack-up for HDQFP172

Due to the nature of HDQFP172 packages, there are no restrictions or requirements on the number of layers used for the HDQFP172 package. However, the recommended stack-up specified for the VFBGA184 package still applies for the HDQFP172.

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4 Bulk and decoupling capacitance

Bulk and decoupling capacitors must be used for proper chip operation. Bulk capacitors are used for local power supply to the pin. Decoupling capacitors are used to prevent noise from entering the MCU by shortening it to the ground. Selecting decoupling capacitors that exhibit low reactance in the frequency ranges of interest is essential.

In general, NXP recommends that each power pin has one decoupling capacitor of 0.1 µF and each power domain has one bulk capacitor of 1 µF. The decoupling and bulk capacitors must be placed as close to their respective MCU pins as possible. Table 2 presents the minimum recommended bulk and decoupling capacitors. These values are based on the assumption that no devices operate in the system with a higher operating frequency than the MCX device.

Table 2. Minimum recommended bulk and decoupling capacitors

Power domain	VFBGA184 / HDQFP172 recommendations	HLQFP100 recommendations	Remarks
VDD_LDO_CORE	**LDO configuration:** 1x 0.1 µF **DCDC configuration:** N/A	N/A [1] VDD_LDO_CORE is shared with the VDD pin.	If the core LDO is not used, omit this capacitor. It is also not applicable for the HLQFP100 package.
VDD_CORE	**LDO configuration:** x 3.7 µF - 10 µF (4.7 µF is recommended) 2 x 0.1 µF **DCDC configuration:** 1 x 6 µF - 30 µF 2 x 0.1 µF	**LDO configuration:** 1x 3.7 µF - 10 µF (4.7 µF is recommended) 2 x 0.1 µF **DCDC configuration:** 1 x 6 µF - 30 µF 2 x 0.1 µF	Each VDD_CORE pin must have a 0.1 µF capacitor near to it. The larger bulk capacitance must be placed as near as possible. However, only one capacitor is necessary for all VDD_CORE pins on the device package.
VDD_LDO_SYS	1x 1.0 µF	VDD_LDO_SYS is shared with the VDD_DCDC pin.	If the LDO system is not used, omit this capacitor. It is also not applicable for the HLQFP100 package.
VDD_SYS	1x 1.0 µF to 4.0 µF	1x 1.4 µF to 4.0 µF	2.2 µF is the recommended value.
VDD	1x 1 µF and 3x 0.1 µF	1x 1 µF and 3x 0.1 µF	-
VDD_P2	1x 1 µF and 2x 0.1 µF	N/A [2] VDD_P2 is shared with VDD in this package.	This row is not applicable to the HLQFP100 package.
VDD_P3	1x 1 µF and 3x 0.1 µF	1x 1 µF 3x 0.1 µF	-
VDD_P4	1x 1 µF and 2x 0.1 µF	1x 1 µF 1x 0.1 µF	-
VDD_ANA	-	-	It must be tied to VDD_P4 through a filtering inductor (which is optional). As such, this net uses VDD_P4 bulk and decoupling capacitance.
VDD_BAT	1x 0.1 µF	1x 0.1 µF	-
VDD_USB	1x 0.1 µF	1x 0.1 µF	If the USB is not used, omit this capacitor.
VREFH	1x 1 µF	1x 1 µF	-

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Table 2. Minimum recommended bulk and decoupling capacitors*...continued*

Power domain	VFBGA184 / HDQFP172 recommendations	HLQFP100 recommendations	Remarks
VDD_DCDC	**LDO configuration:** NA **DCDC configuration:** 1x 10 µF	**LDO configuration:** 1x 1.0 µF **DCDC configuration:** 1x 10 µF	If the DCDC is not used, this capacitor can be omitted. Otherwise, the 10 µF capacitor must be placed on the input to the DCDC.

[1] There is no VDD_LDO_CORE pin.
[2] There is no VDD_P2 pin.

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5 Power configurations

Two basic power configurations are allowed for the MCX N series of MCUs:

- Single supply configuration
- Mixed supply configuration

5.1 Single supply configuration

This configuration is a straightforward configuration where a single power source between 1.71 V - 3.6 V supplies power to the entire system. The required power sequencing is inherently adhered to by the nature of the configuration. The schematic drawings provided in the previous sections of this document are drawn using a single supply.

Important:

In the single supply configuration, sources less than 2.75 V cannot blow fuses. It must be considered in the design phase of your project. Supplies of 1.95 V or less must power the `VOUT_SYS` pin directly and disable the LDO_SYS regulator.

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5.2 Mixed supply configuration

In this configuration, two or more voltage supplies power up the system. Both must be in the range of 1.71 V to 3.6 V and power sequencing rules, see Figure 15.

Note: *Die pins are shown in the figure below. Ensure to check your package-specific pinout, as some packages double-bond these die pins to package pins.*

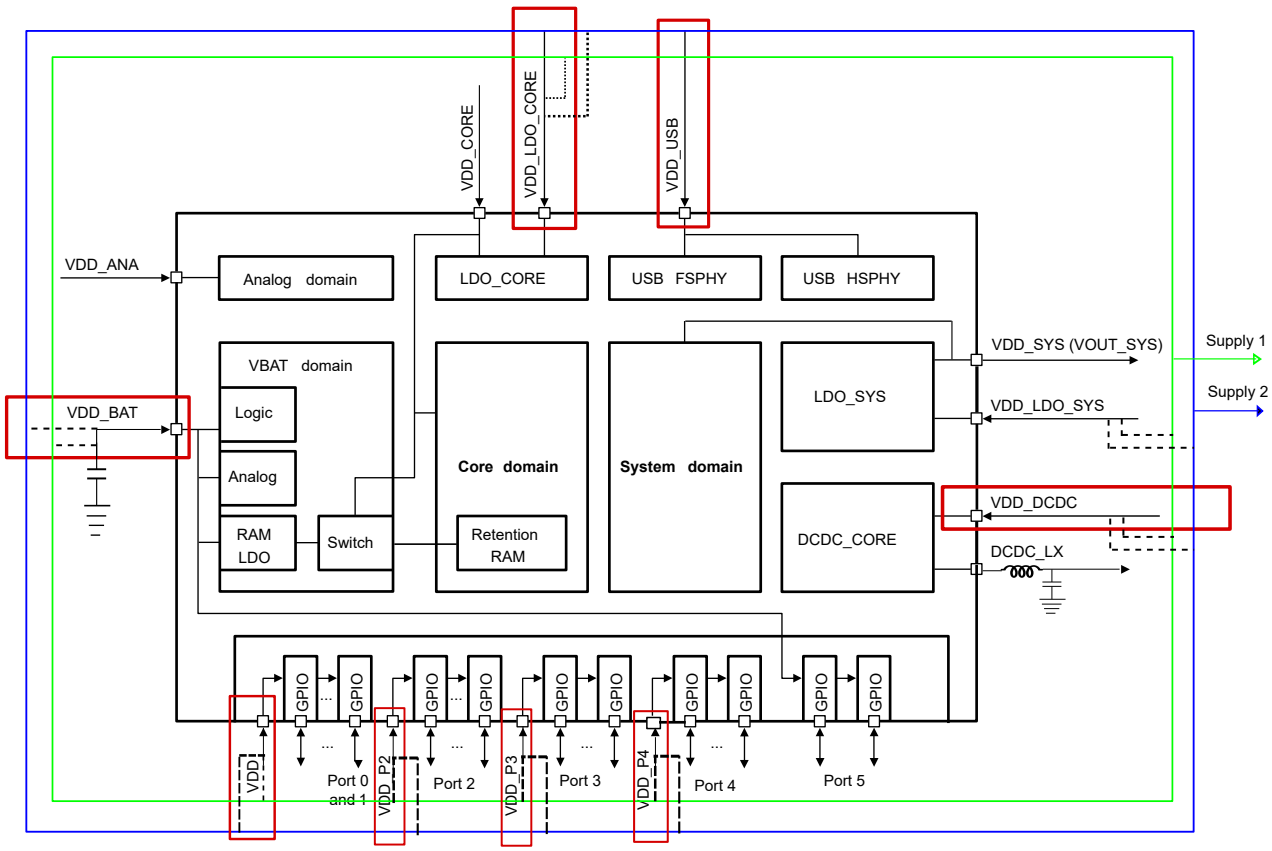


Figure 15. Mixed supply power configuration

The power sequencing rules that must be followed are:

1. Secondary IO supplies (VDD_P2, VDD_P3, VDD_P4) must implement one of the following:
   - Must be short with VDD.
   - Or must ramp after VDD_SYS.
2. VDD_CORE must ramp after VDD.
3. VDD_P4 and VDD_ANA must be at the same voltage level.
4. VDD_BAT must ramp before or with VDD_SYS.

There are too many possible scenarios to cover each one in detail in this guide. If the below mentioned domains are powered by the rail that ramps first, general power sequencing issues can be avoided:

- Core domain (either VDD_LDO_CORE or VDD_DCDC)
- VDD_LDO_SYS
- VDD
- VDD_BAT

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Note: *If the rail that ramps first is below the minimum voltage of 1.95 V (or 2.75 V if fuse programming is needed) of the SYS_LDO, then the higher of the two voltages must power up this regulator.*

For best power consumption:

1. If the Core LDO powers up the core domain, use the lower of the voltages to power the Core LDO input. Follow this guideline unless it causes one of the above rules to be broken.
2. If the DCDC powers up the core domain, use the higher of the two voltages to power the DCDC. Follow this guideline unless it causes one of the above rules to be broken.

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6 DCDC

The MCX N series includes an on-chip DCDC converter to regulate the VDD_CORE domain voltage. The DCDC regulator is required to achieve optimal power performance of the MCX N series devices. An external inductor is needed for the use of the DCDC converter. The inductor must be placed between the DCDC_LX pin and the VDD_CORE pin. It must meet all the specifications described in the *Section 'DCDC converter specifications'* of the MCX N data sheet. (Refer to the appropriate data sheet for your device.) An unshielded inductor can be used, though using it is not recommended due to emissions and performance reasons.

6.1 DCDC inductor layout and routing

Proper routing and layout of the DCDC inductor are essential for optimal DCDC performance and for avoiding unwanted emissions from the inductor. When routing the DCDC inductor, the following rules must be observed:

1. The loop must be on the same layer as the MCX N series device.
2. The input capacitor must be closely connected to the output capacitor and input pin.
3. The ground connection trace from the input capacitor to the load capacitor must be well stitched.

An example DCDC circuit layout for the BGA package is shown in Figure 16.

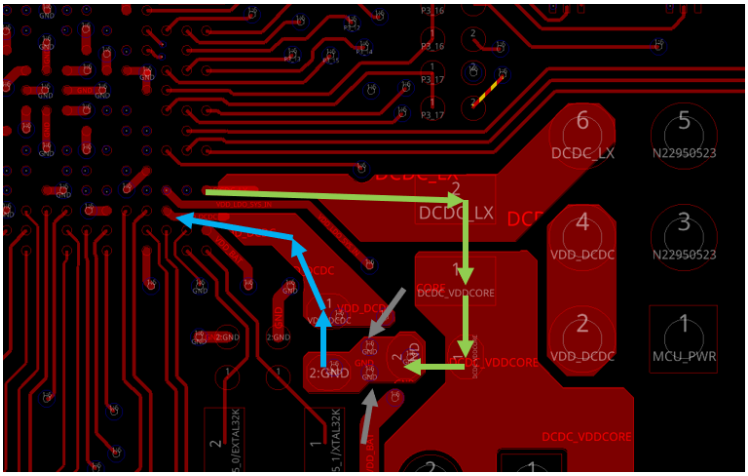


Figure 16. Optimal DCDC routing for BGA packages

The loop referred to in the layout rule 1 refers to the path that the current follows during the charging and discharging phases of the DCDC. During the charging phase, the current starts from the DCDC_LX pin and travels through the inductor and then through the load capacitor to the ground. The yellow arrows in Figure 16 highlight the current path. During the discharging phase, the current is pulled from the ground, through the bulk capacitor, and to the VDD_DCDC pin. The blue arrows shown in Figure 16 indicate this return path. This entire loop path must be minimized for optimal DCDC operation.

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Figure 17 shows the optimal DCDC routing for HLQFP packages.

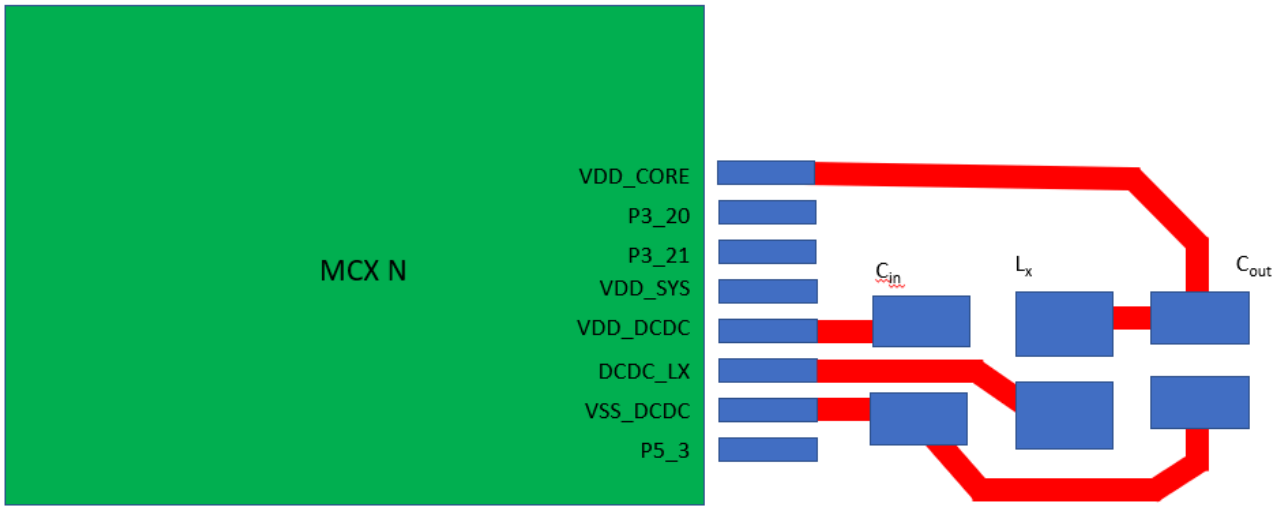


Figure 17. Optimal DCDC routing for HLQFP package

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Figure 18 shows the optimal DCDC routing for the HDQFP package.

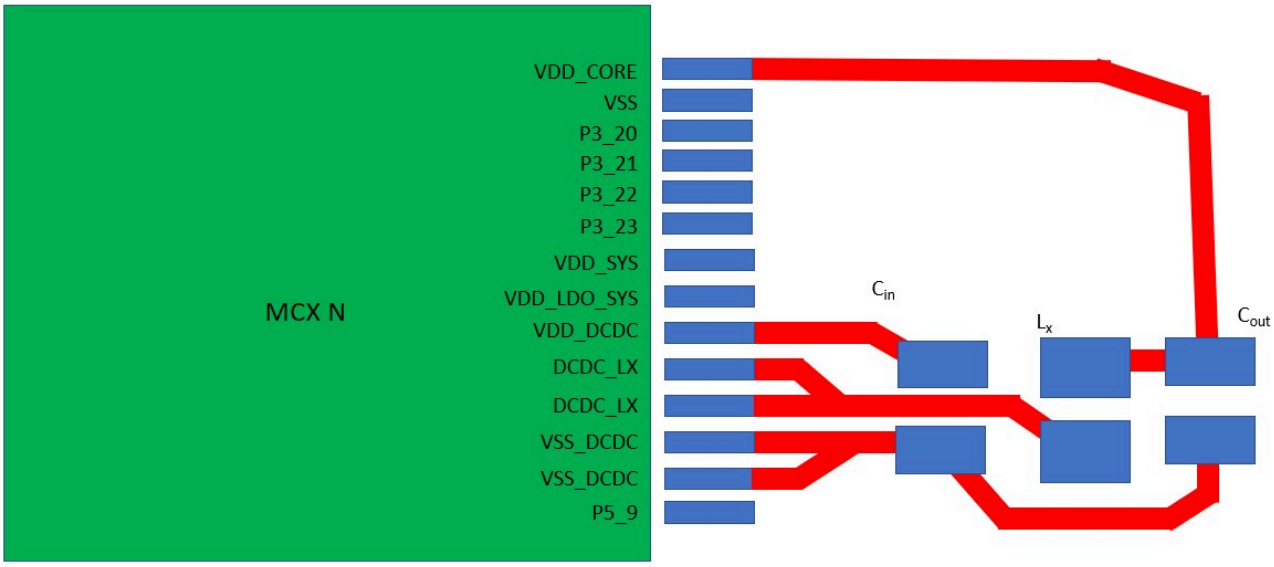


Figure 18. Optimal DCDC routing for the HDQFP package

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7 Clocks

The MCX N series of MCUs include two external oscillator modules: a 32 kHz oscillator module and a high frequency 16 MHz – 50 MHz oscillator module. The layout and routing of the modules are the same, but there are some differences as follows:

- The 32 kHz oscillator module has an internal capacitor bank capable of 0 pF to 30 pF in steps, so external load capacitors are not required. Therefore, the load capacitors are marked DNP.
- The high-frequency oscillator has a high-gain mode resistor footprint. The high-frequency oscillator does not have an internal capacitor bank. Therefore, external load capacitors are required.

The schematics for the two oscillator modules are shown in Figure 19 and Figure 20.

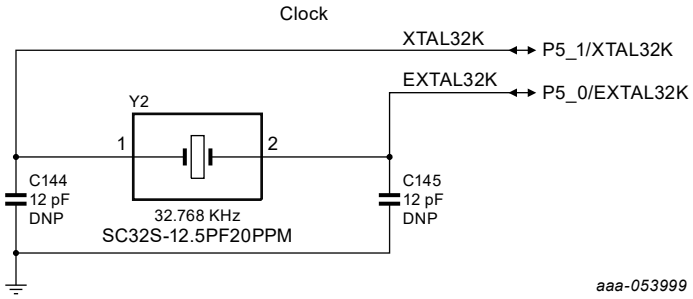


Figure 19. 32 kHz oscillator schematic

Note: *C144 and C145 are marked DNP in Figure 19.*

If needed, the case includes footprints for the capacitors.

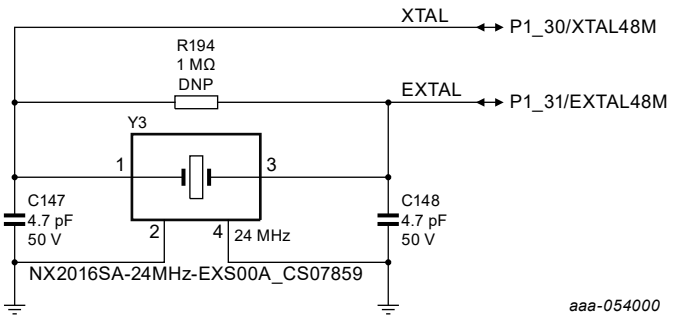


Figure 20. High-frequency oscillator schematic

For more details, see Figure 21.

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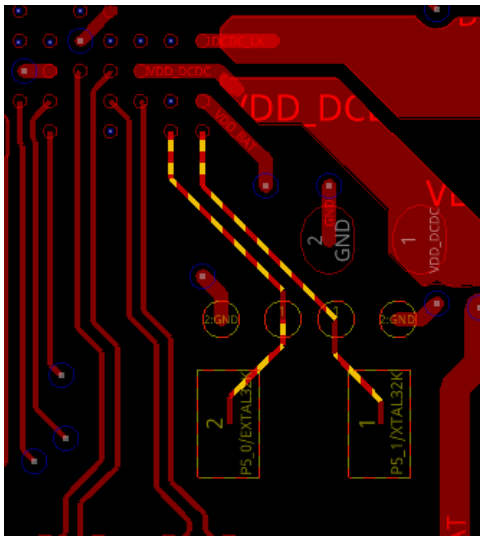


Figure 21. Recommended oscillator layout

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8 Digital interfaces

This section provides details about general high-speed design recommendations and debug circuits. It also includes the guidelines for using interfaces such as high-speed USB, full-speed USB, Ethernet, uSHDC, and FlexSPI.

8.1 General high-speed design recommendations

NXP recommends using high-speed design techniques for most of the digital interfaces. While the interface speeds are only up to 100 MHz, and some interfaces are slow enough that these recommendations do not need to be strictly followed. These techniques are good practice and are not cost-prohibitive recommendations. The recommended layout techniques are detailed below and apply to the data and signaling lines of all digital interfaces unless otherwise specified:

- The route traces over continuous planes.
  - Do not route over power/ground plane slots or antietch.
- Length matches data and signaling lines.
- Do not route data and signaling lines under oscillators or parallel to other clock traces and/or data buses.
- Minimize the number of corners needed. When corners are needed, use 45 degree turns instead of 90 degree turns.
- Minimize data and signaling line length.
- Avoid layer changes.
- Do not create stubs or branches.
- Provide the ground return vias within a 50 mil distance from the signals that layer-transition vias when transitioning between different reference ground planes.

8.2 Debug circuit

NXP recommends a pull-up resistor on the `SWDIO/TMS` signal and a pull-down resistor on the `SWDCLK/TCK` signal. Other pull-ups are not required. The debug pins are port `P0` pins, so they must be pulled up to VDD. Figure 22 shows the schematic except for the MCX-N9XX-EVK.

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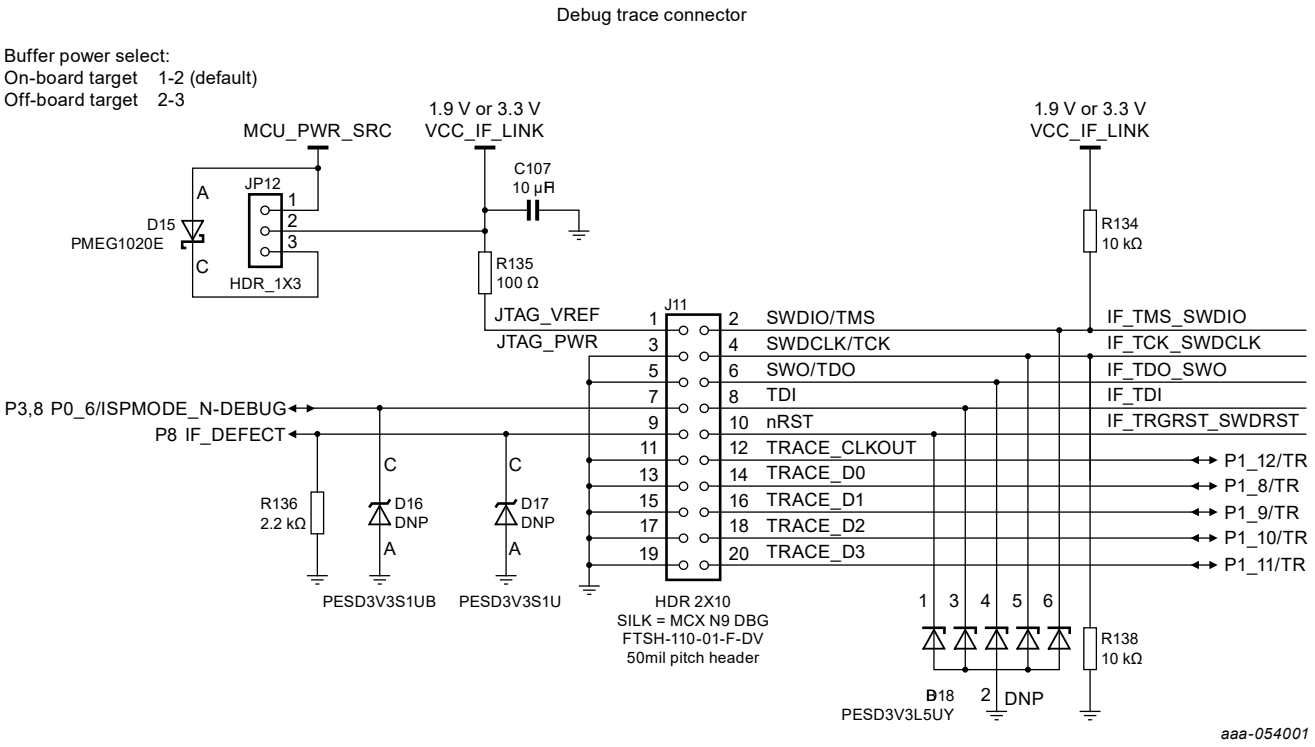


Figure 22. Example debug circuit schematic

Note: *`MCU_PWR_SRC` sources to VDD are not shown Figure 22, so the JTAG pins can only be pulled up to VDD levels.*

- The MCX-N9XX-EVK also includes ESD diodes for the critical pin connections. It is recommended but optional.
- The trace signals `TRACE_CLKOUT`, `TRACE_D0`, `TRACE_D1`, `TRACE_D2`, and `TRACE_D3` must be treated as the data/signaling lines and must be routed with the recommendations detailed in Section 8.1 "General high-speed design recommendations".

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8.3 High-speed USB

The USB controller block provides high-performance Universal Serial Bus (USB) functionality that conforms to the USB Specification Rev. 2.0 (Compaq, Hewlett-Packard, Intel, Lucent, Microsoft, NEC, Philips; 2000). The controller is also compliant with the On-The-Go and embedded host supplement to the USB revision 2.0 specification (Hewlett-Packard Company, Intel Corporation, LSI Corporation, Microsoft Corporation, Renesas Electronics Corporation, ST-Ericsson; 2012). There are four dedicated pins to consider in your hardware design:

- USB D+
- USB D-
- USB ID
- USB VBUS

The USB D+ and D- signals must be treated as the data/signaling lines and must be routed with the recommendations detailed in Section 8.1 "General high-speed design recommendations".

The USB routing must follow these recommendations:

- Route the D+ and D- differential pairs first.
- Route the D+ and D- signals on the top or bottom layer of the board.
- The trace width and spacing of the D+ and D- signals must meet the differential impedance requirement of 90 Ω.
- When placing the connectors, make sure that the ground plane clearouts around each pin have ground continuity between all pins.
- Maintain parallelism (skew-matched) between D+ and D-, and match the overall differential length difference to fewer than 5 mil.
- Maintain symmetric routing for each differential pair.
- Minimize the lengths of the high-speed signals that run parallel to the D+ and D- pair.
- These pins are not required and must not use a termination resistor. They must be routed directly to the USB connector or to any Common mode chokes used in the design.
- The use of TVS diodes on these dedicated pins is also recommended.

The USB VBUS pin is a unique pin in that it is the only 5 V tolerant pin. The VBUS pin from the connector must be routed directly to the `USB_VBUS` MCX pin.

The High-speed USB (HS USB) module requires the HS USB PHY module, which requires the use of the USB PLL. This module requires specific crystal frequencies to be used. Therefore, if your application uses HS USB, it must use the high-frequency external oscillator. Use one of these specific crystal frequencies:

- 12 MHz
- 16 MHz
- 19.2 MHz
- 20 MHz
- 24 MHz
- 30 MHz
- 32 MHz
- 38.4 MHz
- 40 MHz
- 48 MHz

If using the HS USB in In-system programming (ISP) mode, a 24 MHz crystal must be used in at least the first access of the device. However, the user can modify the security fuse configuration to allow a different crystal. However, the user must configure it. For example, factory configuration requires a 24 MHz crystal for HS USB ISP operation.

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8.4 Full-speed USB

The USB controller block provides high-performance USB functionality that conforms to the Universal Serial Bus Specification, Rev. 2.0 (Compaq, Hewlett-Packard, Intel, Lucent, Microsoft, NEC, Philips; 2000). It also conforms to the On-The-Go and Embedded Host Supplement to the USB Revision 2.0 Specification (Hewlett-Packard Company, Intel Corporation, LSI Corporation, Microsoft Corporation, Renesas Electronics Corporation, ST-Ericsson; 2012).

There are only two dedicated pins to consider in your hardware design:

- USB D+
- USB D-

The USB VBUS detect pin for the FS USB block is not 5 V tolerant. A voltage divider or other similar translator circuit is necessary for proper operation. When designing the USB circuit, it is important to know if you plan on implementing a crystal-less design. Crystal-less designs are only supported in Device mode. Other modes require the high-frequency oscillator to use a crystal of specific frequencies.

The USB D+ and D- signals must be treated as the data/signaling lines. While routing, adhere to these signals, follow the recommendations detailed in Section 8.1 "General high-speed design recommendations".

The USB routing must follow these recommendations:

- Route the D+ and D- differential pairs first.
- Route the D+ and D- signals on the top (or bottom) layer of the board.
- The trace width and spacing of the D+ and D- signals must meet the differential impedance requirement of 90 Ω.
- When placing the connectors, ensure that the ground plane clearouts around each pin have ground continuity between all pins.
- Maintain parallelism (skew-matched) between D+ and D-, and match the overall differential length difference to fewer than 5 mil.
- Maintain symmetric routing for each differential pair.
- Minimize the lengths of the high-speed signals that run parallel to the D+ and D- pair.
- Provide the ground return vias within a 50 mil distance from the signals that layer-transition vias when transitioning between different reference ground planes.
- 33 Ω termination resistors are required for proper operation.
- TVS diodes are strongly recommended, and common mode chokes are optional.

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8.5 Ethernet

The Ethernet module enables a host to transmit and receive data over Ethernet in compliance with the IEEE 802.3-2008 standard. The Ethernet interface contains a full-featured 10 Mbit/s or 100 Mbit/s Ethernet media access controller (MAC) designed to provide optimized performance by using DMA hardware acceleration.

Features include the following:

- Full-duplex or half-duplex operation
- RMII and MII support
- Flow control and control frames
- Hardware acceleration for transmit retry
- Receive packet filtering
- Wake-up on LAN activity supporting both wake-up and magic packet frames.

8.5.1 RMII

This section provides information about Reduced Media Independent Interface (RMII), Ethernet schematic, and layout recommendations.

8.5.1.1 Ethernet schematic recommendations

Sourcing the clock for an Ethernet PHY is an important aspect of implementing the Ethernet in a design. The reason is the specific timing requirements of the Ethernet standard. Either the MCU or the PHY can supply the Ethernet clock. The option that you select depends on the selected PHY and individual situations. MCX N supports either of these configurations via the same pin.

Figure 23 shows the Ethernet clocking scheme.

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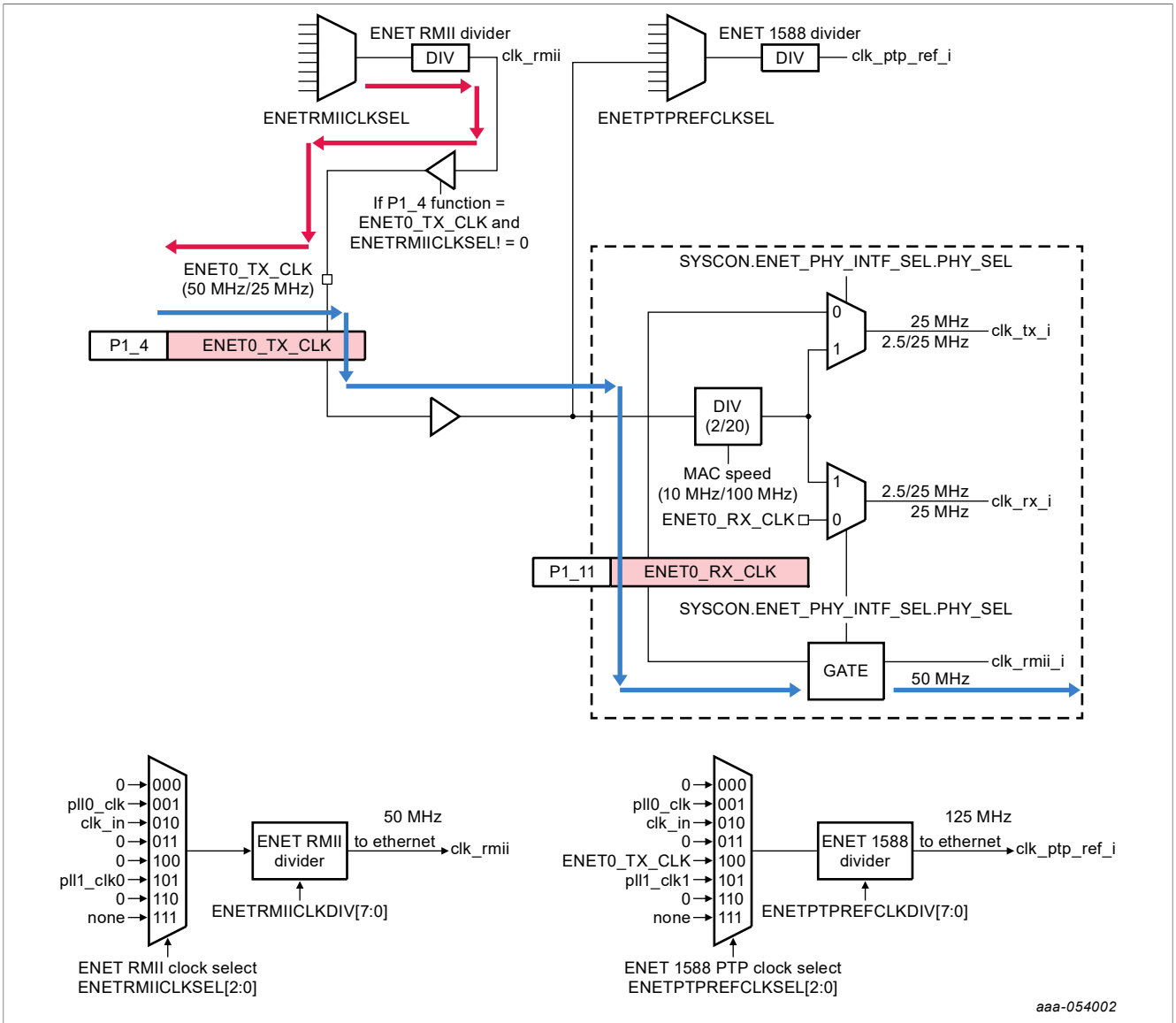


Figure 23. Ethernet clocking scheme

As shown by the red arrows in Figure 23, the Ethernet clock is output to the PHY through the `ENET0_TX_CLK (P1_4)` when software configures `P1_4` to select the `ENET0_TX_CLK` mux function and `ENETRMIICLKSEL` is not `0x0`. As shown by the blue arrows in Figure 23, pin `P1_4` accepts the RMII clock from the PHY when the `P1_4` digital input buffer is enabled `ENET_PHY_INTF_SEL[PHY_SEL]` selects the RMII clock.

In addition to the previous recommendation, NXP also recommends impedance matching the MCU to PHY signals, for which a termination resistor is often required. It is important to consult the design tools to determine if the resistor is necessary.

8.5.2 Ethernet layout recommendations

Given the relatively high-speed and tight timing requirements, NXP recommends using high-speed design techniques when routing the Ethernet in MII or RMII configuration. The `RXDx`, `TXDx`, `RXDV` (RMII only), `TXEN` (RMII only), `RXCLK` (MII only), `TXCLK` (MII only), `TXER` (MII only), `RXER` (MII only), `CRS` (MII only), and `COL` (MII only) signals must be treated as the data/signaling lines. While routing these signals, follow the recommendations detailed in Section 8.1 "General high-speed design recommendations".

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8.6 uSHDC

The uSDHC module provides the interface between the host system and the eMMC, SD card, and SDIO media. The module acts as a bridge between the eMMC, SD card, and SDIO. It allows host bus transactions, sends commands, and performs data accesses to or from the cards. It handles the SD card, SDIO, or eMMC protocols at the transmission level. It supports up to 104 MHz interface frequencies in SDR mode and 50 MHz in DDR mode. The data/signaling lines must be routed with the recommendations detailed in the Section 8.1 "General high-speed design recommendations". The following signals must be considered for data/signaling lines:

- CLK
- CMD
- DATx

Note: *The MCX uSDHC module does support using DAT3 as the card detection line.*

For details on how to build SD-memory card and MMC card systems with EMI filtering and protection compliant with the relevant standards, refer to document AN10911. (See https://www.mouser.com/catalog/specsheets/an10911.pdf).

8.7 FlexSPI

FlexSPI is a flexible SPI host controller that supports two SPI channels and up to four external devices. Each channel supports Single/Dual/Quad/Octal mode data transfer and 1/2/4/8 bidirectional data lines. FlexSPI is most commonly used to interface to external memory.

For more information, refer to the Section 'FlexSPI specifications' in the MCX N series Data Sheet (document MCXNP184M150F70). There are several sources for the internal sample clock for FlexSPI read data:

- Dummy read strobe generated by the FlexSPI controller and looped back internally `FlexSPIn_MCR0[RXCLKSRC] = 0x0`.
- Dummy read strobe generated by the FlexSPI controller and looped back through DQS pad `FlexSPIn_MCR0[RXCLKSRC] = 0x1`.
- SCLK output clock and looped back from SCLK pad `FlexSPIn_MCR0[RXCLKSRC] = 0x2`.
- Read strobe provided by a memory device and input from a DQS pad `FlexSPIn_MCR0[RXCLKSRC] = 0x3`.

For QSPI Flash without a DQS provided by the memory, only the option of `FlexSPIn_MCR0[RXCLKSRC] = 0x1` can achieve 100 MHz SDR R/W speed, and the `FlexSPI_DQS` pin must be left floating.

The Octal flash, where a DQS signal is provided by the memory, must use the option of `FlexSPIn_MCR0[RXCLKSRC] = 0x3`, which can achieve 166 MHz DDR R/W. In such a case, the `FlexSPI_DQS` pin must be connected to the flash directly.

In addition, given the high-speed capabilities of the FlexSPI interface, high-speed design techniques must be incorporated. Treat the SSx, SCLK, DQS, and DATAx signals as data/signaling lines. Route these signals adhering to the recommendations detailed in Section 8.1 "General high-speed design recommendations".

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9 Analog interfaces

This section provides details about ADC, DAC, and VREF.

9.1 ADC

The conversion accuracy of an ADC significantly depends on two factors:

- The sampling circuit connected to the ADC channel
- The quality of the input to the ADC reference (VREFH / VREFL).

This section first discusses the ADC reference input (VREFH / VREFL).

9.1.1 ADC reference input

The ADC reference input has minimal power supply noise rejection. The inductance of the trace from the ADC reference source to the ADC reference input, VREFH / VREFL is important. To ensure maximum noise rejection for the ADC reference, consider this trace inductance.

If the inductance of the trace from the ADC reference source to the ADC reference input VREFH / VREFL exceeds 0.5 nH of inductance, route the VREFH / VREFL as a differential pair. The two examples shown in Figure 24 and Figure 25 display these cases.

Example #1: Solid singular ground plane underneath with ADC reference close to MCX device

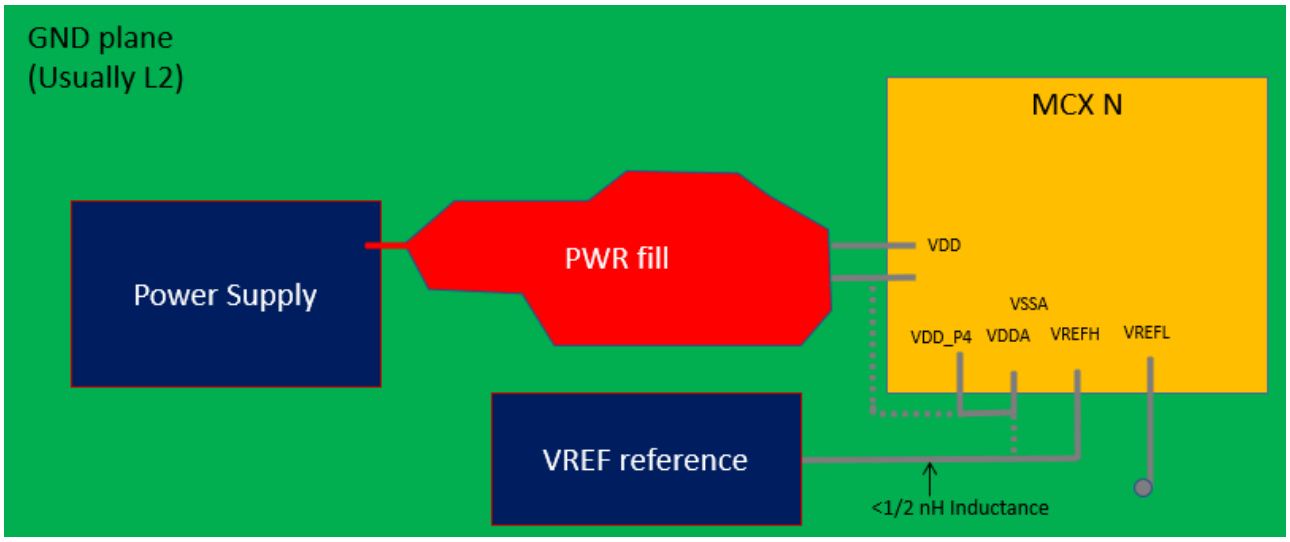


Figure 24. Solid singular ground plane example

In this case, the trace from the VREF reference and VREFH is less than ½ nH of inductance. VREFL can then be tied to GND underneath the MCX device or the nearest place to connect to the GND plane.

Example #2: Split ground plane with ADC reference significantly far from MCX device

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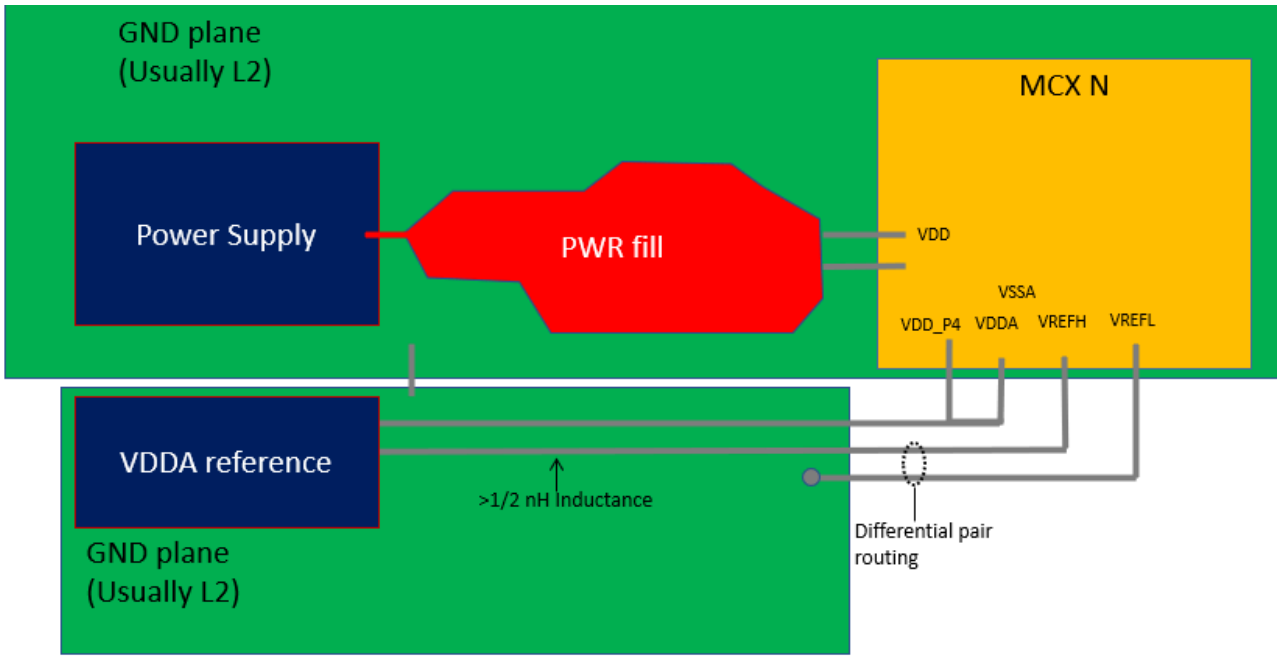


Figure 25. Split ground plane example

In this case, not only is the reference for VREFH significantly far away from the MCX device, there is not a well-defined ground plane beneath the VREFH / VREFL route. Therefore, the VREFH / VREFL pair must be routed as a differential pair at least until the VREFL trace can be tied to a well-defined quiet ground plane.

To determine the inductance of a trace, most PCB design software provides a method for estimating this value. Refer to the user manual of your software for instructions on how to make this estimation.

9.1.2 ADC sampling input

To achieve optimal conversion accuracy, a sampling circuit must use a buffered input. If the end application requires the best ADC accuracy and a high sampling frequency, using a buffered input is especially important. However, for applications that cannot implement a buffer or have some tolerance in the ADC performance, a simple RC filter input circuit is still acceptable.

With any design, the connected circuit must still meet the sample time requirements specified in the below equation:

$$T_{SMP_REQ} = B*Ln(2)*\left[R_{AS}*\left(C_{AS}+C_P+C_{ADIN}\right)+\left(R_{AS}+R_{ADIN}\right)*C_{ADIN}\right] \quad (1)$$

Where:

- T_{SMP_REQ} is the sample time for which the ADC requires to wait while sampling the channel.
- B is the desired accuracy in bits (for example, to acquire the input to within ½ LSB of 12 bits, select B = 13).
- R_{AS} is the total effective series resistance of the input channel external to the MCX device, including driver circuit and any external RC filter series resistance.
- C_{AS} is the total external capacitance on the input channel external to the MCX device, including any external RC filter capacitance.
- C_P is the capacitance of the pin (specified in the data sheet).
- C_{ADIN} is the ADC capacitance (specified in the data sheet).
- R_{ADIN} is the series resistance of the ADC channel used (specified in the data sheet).
- Ln(2) is the natural log of 2.

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Calculate the sample time of the ADC, T_{SMP}, by using the equation below:

$$T_{SMP} = \frac{3.5 + 2^{STS}}{F_{ADCK}} \quad (2)$$

Where,

- STS is the value of the Sample Time Select bit field, CMDHa[STS], in the Command High Buffer Register. Refer to the *MCX N94x / N54x Reference Manual* for more information.
- F_{ADCK} is the frequency of the functional clock to the ADC.

Conversely, if you have a specific sampling time, you must know the R_{AS} and meet the required value. For such cases, solve the equation for R_{AS} and achieve the resulting equation:

$$R_{AS} < \frac{\frac{3.5 + 2^{STS}}{F_{ADCK} * B * ln(2)} - R_{ADIN} * C_{ADIN}}{C_{AS} + C_P + 2 * C_{ADIN}} \quad (3)$$

The above equations assume that the input voltage can change up to full-scale ($V_{REFH} – V_{REFL}$) between subsequent conversions. If the maximum input voltage change between subsequent conversions is known and less than V_{REFH}-V_{REFL}, then the value of B can be adjusted.

For example, for a case for which ½ LSB sample accuracy is 14 bits and the maximum input voltage change is known to be only ($V_{REFH} – V_{REFL}$) / 4, B can be set to 13 rather than 15.

9.2 DAC

The DAC requires a load capacitance to achieve optimal performance. This capacitance must be configured in a standard low-pass filter configuration. This range is specified in the *Section "12-bit DAC operating requirements"* or *"14-bit DAC operating requirements*" (if the 14-bit DAC is being used) of the MCX N Data Sheet. (Refer to the appropriate data sheet for your device listed in Section 13 "References").

9.3 VREF

The VREF requires a load capacitance to achieve optimal performance. The nominal load capacitance required is stated in Table *"VREF operating requirements"* of *MCX N series Data Sheet* (document MCXNP184M150F70). This capacitor must be placed as close to the VREF pin as possible in a standard low-pass filter configuration. It must be included whether using the VREF reference internally or externally.

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10 HMI interfaces

This section provides details about GPIO and TSI interfaces.

10.1 GPIO

There are six GPIO ports on the MCX N series device: PORT0, PORT1, PORT2, PORT3, PORT4, and PORT5. The VDD domain sources ports 0 and 1, while ports 2, 3, 4, and 5 have independent domain voltages. It is important to understand the voltage source that supplies a specific pin. No pin on the device must exceed the source voltage by more than 0.3 V. Likewise, no pin must have a voltage less than ground applied to it by more than -0.3 V. Designs must be reviewed carefully to ensure that neither of these restrictions is violated.

In addition, there are restrictions on the amount of current that individual pins can source and limits to the ports collectively. Individual pins cannot source or sink more than +/- 25 mA, and the sum of the currents of all pins on a port cannot exceed 100 mA.

Pads are categorized into fast, medium, and slow speed pads. Each pad type has its own characteristics, which are specified in *MCXN Data Sheet* (refer to the appropriate data sheet for your device listed in Section 13 "References".). The pad type can be found in the pinout spreadsheet and table, see Figure 26.

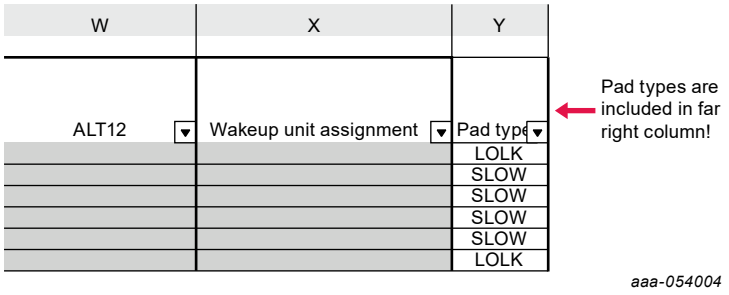


Figure 26. Pad type locations in pinout spreadsheet

10.1.1 Pull-up and pull-down resistors

All pads on the MCX N series devices support internal pull-up or pull-down resistors. These resistors are weak resistor with a value of typically 50 kΩ. Therefore, they are not recommended for use with some applications, for example, I^2C.

Some pins also support a higher pull-up or pull-down resistance option to support lower leakage applications. The pins that support this option are listed in the PCR register descriptions in the Section *"PORT register descriptions"* in the *MCX N Data Sheet*. (Refer to the appropriate data sheet for your device.) See Section 13 "References".

10.1.2 Reset pin

The reset pin on this device is a bidirectional, open-drain dedicated pin with an internal pull-up resistor. The one point that must not be overlooked from a hardware design perspective is that a 100 nF bypass filtering capacitor must be placed near this pin. The capacitor must be connected in a typical low-pass filter configuration, reset pin to the ground. This capacitor is a requirement.

Figure 27 is an example of the circuit for the required capacitor attached to the reset pin.

Note: *The net `MCU_RESET_B` is directly connected to the reset pin of the MCX device.*

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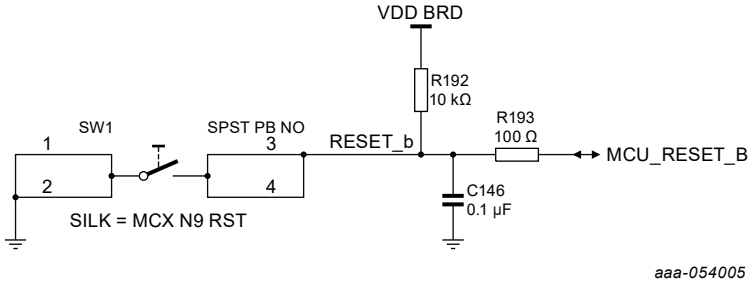


Figure 27. Reset pin example circuit

10.1.3 ISP pin

The ISP mode supports image programming via the serial interface. One way to enter this mode is by holding the ISP pin `P0_6` low during the reset process. After this mode is entered, debug communication is not possible. The user application cannot run until a valid ISP interface successfully commands the device to run the user application. Therefore, it is important that hardware designs appropriately handle this pin. How this pin is handled is application-dependent, but hardware designs must ensure that the use of this pin is understood. The pin must not be inadvertently pulled down as this causes the device to enter ISP mode. The internal pull-up of the pin is enabled by default. Therefore, use external pull-ups only if external factors dictate that it is necessary.

If a design uses ISP mode, all interfaces planned to be used must be made available as appropriate. The pins/module instances used for ISP interfacing can be changed. However, it is important to note that the default pin/module combination must be used on the first access of a new device. Therefore, designs that use ISP mode in manufacturing must use at least one of the default pin/module combinations.

The default ISP mode pins/module instances are mentioned in the pinout spreadsheet attached to the *MCX N Reference Manual* (refer to the appropriate data sheet for your device). Alternatively, they can be found in the "*Pinout*" Table of the *MCX N Reference Manual*. See Section 13 "References".

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10.2 TSI

The Touch System Interface (TSI) provides touch sensing detection of capacitive touch sensors. The external capacitive touch sensor is typically formed on a PCB, and the sensor electrodes are connected to TSI input channels through the I/O pins. The TSI module operates in a switching-integration style to achieve low-power, high-sensitivity, and advanced Electromagnetic Compatibility (EMC) robustness.

10.2.1 TSI pin selection

The TSI module supports two types of modes:

- Self-capacitance
- Mutual-capacitance

A different pattern is required for each mode. Any TSI capable pin is capable of the Self-capacitance mode. However, because the Mutual-capacitance mode has a TX and RX pin, care must be taken to ensure that the TX and RX are correctly selected and connected. TX pins are TSI channels 0 - 7, and RX pins are TSI channels 8 - 24.

10.2.2 Electrode design for Self-capacitance mode

Usually, the electrode size is around 5 - 15 mm. A larger size of the electrode is appropriate for thicker overlays. To maximize the area of the electrodes from the capacitor plates, it is recommended that the electrode size be similar to a human finger, as 10 × 10 mm is considered a good size. To prevent charges from accumulating at the tips, try to avoid sharp corners when designing touch electrodes, see Figure 28.



Figure 28. Electrode shapes for Self-capacitance mode

10.2.3 Electrode design for Mutual-capacitance mode

A mutual key is used to connect the TX and RX channels of TSI. It detects whether the key is pressed by sensing the change of mutual inductance capacitance between the RX and TX. The electric field between RX and TX must be affected to the greatest extent when the finger touches the key. This check must be ensured while designing the pattern of the mutual key.

The recommended mutual key shape is shown in Figure 29. The electrode of the TX wraps the electrode of the RX, which prevents noise from affecting the RX channel. The number of fingers has a great impact on touch sensitivity. In general, more fingers result in stronger noise immunity but less touch sensitivity. Customers must select the correct number of fingers for the thickness of the touch overlay. For example, if the touch overlay is 3 mm thick, four fingers is the best choice. If the touch overlay is 2 mm thick, five fingers are sufficient.

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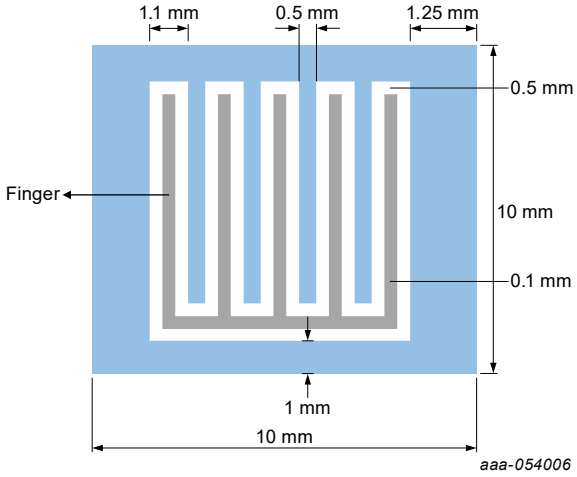


Figure 29. Electrode shape for Mutual-capacitance mode

10.2.4 TSI PCB trace routing

The following are recommendations for correctly routing the traces of capacitive electrodes.

- Width: Keep the trace width as thin as possible. A 5 – 7 mil trace width is recommended. A 5 mil trace width has half the capacitive coupling with the planes compared to a 10 mils trace.
- Length: Keep the trace length as short as possible. Trace length must be less than 300 mm. To optimize signal strength, minimize the trace length from the TSI pins to the touch pads.
- Clearance: To ensure signal integrity, leave a minimum clearance of 10 mil for the lines that run parallel to each other in the same layer and route perpendicularly to the ones running in adjacent layers. Good design practice is to keep traces separated by as much as the design allows. At the end of the sensor, where the pitch is typically lower than 10 mil, a Bottleneck mode connection is recommended, as shown in Figure 30. It is an example of maintaining adequate clearance in touch sensing traces.
- Avoid routing under the touch electrode. Do not route traces directly under any touchpad. Avoid electrical noise that could be capacitively coupled to the electrodes.
- There must be no components near the touch electrodes.
- Avoid crossover with other signals.
- For the Mutual-capacitance mode key, keep the TX trace as far as possible from the RX trace.

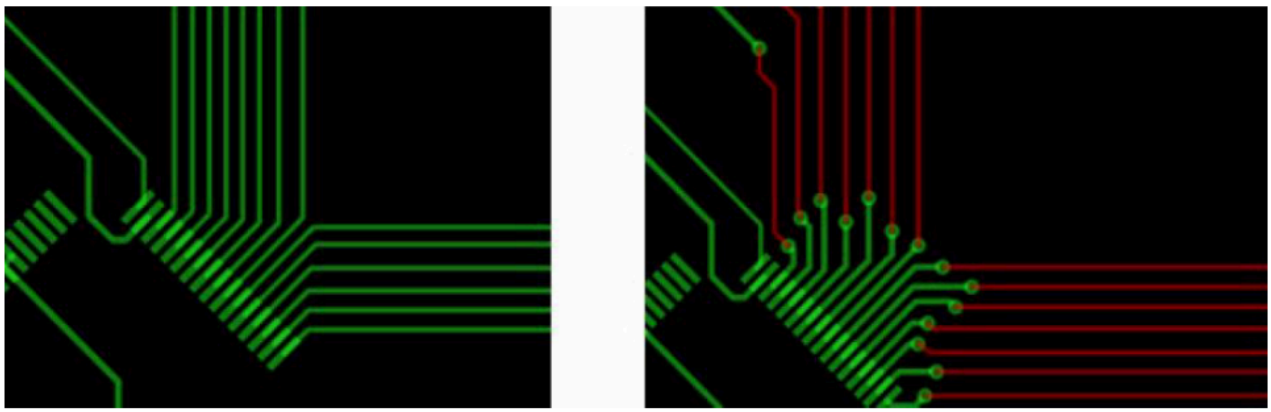


Figure 30. Bottleneck routing maintaining adequate clearance for a touch sensing trace

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10.2.5 TSI ground plane

A proper ground plane prevents the coupling of external electromagnetic interference to the touch-sensing electrodes. It also acts as a shield for undesired electric fields. To use around and under touch electrodes, use an X-hatch pattern ground instead of solid-filled ground. This pattern can decrease the parasitic capacitance and increase the sensitivity of the touch sensor. When there is enough space between electrodes, use an X-hatch pattern between the electrodes. It provides extra noise shielding and reference.

Follow the recommendations and best practices for the ground plane:

- Use X-hatch patterns on the top layer, 25 % ground fill, 7 mil line, 45 mil spacing.
- Use X-hatch patterns on the bottom layer. For example, underneath the electrode area, 17 % ground fill, 7 mil line, 70 mil spacing.

The annular gap size must be equal to the overlay thickness but not smaller than 0.5 mm or larger than 2 mm. For example, a PCB layout for a system with a 1 mm overlay contains a 1 mm annular gap. Another example is a 3 mm overlay design that contains a 2 mm annular gap.

10.2.6 Overlay of touch electrode

To protect the touch electrode from the interference of the external environment, the overlay material must be closely attached to the surface of the touch electrode. There must not be an air gap between the touch electrode and the overlay. Using an overlay with a high dielectric constant or an overlay with a small thickness can improve the sensitivity of the touch electrode.

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11 Unused pins

Table 3 lists the recommendations of unused pins.

Table 3. Unused pin recommendations

Module	Pin/pad types	Recommendation
Analog input pins	ANA	Leave unconnected
Digital	Fast, medium, slow	Leave unconnected, ensure PORTx_MUX = 0x0, pull-up or pull-down resistors disabled, input buffer disabled
USB	USBx_DP, USBx_DM, USB1_ID, USB1_VBUS, VDD_USB	Leave unconnected (except for VDD_USB that must be grounded through a 10 kΩ resistor)
Supply	VDD_LDO_CORE, VDD_LDO_SYS	Tie to the respective output, VDD_CORE, and VDD_SYS pin of the regulator
DCDC	VDD_DCDC, DCDC_LX	**For packages with an independent DCDC supply (VFBGA184 and HDQFP172):** Tie to the ground through a 10 kΩ resistor **For packages with a shared DCDC supply (HLQFP100): Power VDD_DCDC (it cannot be unused in this package)** and leave DCDC_LX unconnected (floating).
Analog supply	VDD_ANA	Tie to VDD_P4
Secondary I/O supply	VDD_P2, VDD_P3, VDD_P4	Leave unconnected (**Note: In the HLQFP100 package, VDD_P2 is powered from VDD and cannot be unused**)
Analog reference	VREFH	Tie to VDD_ANA
	VREFL	Tie to VSS_ANA

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12 Design checklists

This section provides the checklists for designs under review.

For more information, see MCX_N_Series_Design_In_checklist_v4.xlsx.

12.1 Schematic checklist

The tables in this section provide the checklists for schematic designs under review.

Table 4. General checks

S. No.	Check description	Check done (Y/N)	Remarks/Actions
1	Have the power supply requirements and strategy been discussed?	☐	
2	Have the schematics been checked with NXP reference schematics and application notes?	☐	
3	Have several people reviewed the schematics?	☐	
4	Does the application use standard components (components that have been previously used for such applications)?	☐	
5	Have any nonstandard components been qualified so that they can be used in the application?	☐	
6	Are recommendations for layout written on the schematics?	☐	
7	Are the components sized for the desired current drive capability?	☐	
8	Has the MCX symbol pinning been checked?	☐	
9	Have the programming and debug method been discussed?	☐	

Table 5. Oscillator checks

S. No.	Check description	Check done (Y/N)	Remarks/Actions
HF reference oscillator (Note: If USB ISP mode is required with the default configuration, a 24 MHz oscillator is required)			
1	Does the XTAL follow the recommended characteristics?	☐	
2	Are external load capacitors appropriate and included?	☐	
RTC (32 kHz) crystal oscillator			
3	Does the crystal oscillator follow the recommended characteristics?	☐	
4	Are external load capacitors omitted?	☐	

Table 6. Power supply checklist

S. No.	Check description	Check done (Y/ N)	Remarks / Actions
1	Have all the VDD pins been connected according to NXP recommendations?	☐	
2	Are all GND connections connected to GND according to NXP recommendations. If using the HLQFP100 package, has the thermal pad been connected to ground?	☐	
3	Are the power supply regulators well sized?	☐	

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Table 6. Power supply checklist*...continued*

S. No.	Check description	Check done (Y/ N)	Remarks / Actions
4	Has the decoupling of the supply regulators output been implemented?	☐	
5	Have the power supply pins of the IC been properly decoupled (according to the Application Note and Reference Design schematics?)	☐	
6	If the DCDC is used, has an appropriate inductor been sourced and included?	☐	
7	If the DCDC is used, has the core LDO been disabled properly? (if applicable)	☐	
8	If the Core LDO is to be used, has the DCDC been disabled properly? (if applicable)	☐	
9	Do the output capacitors of the onboard regulators (LDO_SYS, LDO_CORE / DCDC) meet data sheet specifications?	☐	
10	Has it been verified that the power sequencing requirements will not be violated?	☐	
11	Have Vrefh / Vrefl been properly constrained within their specified operating limits?	☐	
12	Has proper filtering of VDDA been applied such that the maximum allowable ripple will not be exceeded?	☐	

Table 7. GPIO checks

S. No.	Check description	Check done (Y/ N)	Remarks / Actions
1	Has the compatibility of the I/O logic levels of the MCX with other components been checked?	☐	
2	Does the max source/sink current of the GPIO fit the application?	☐	
3	If pull-ups are used on any GPIO in the design, are the pull-ups sourced from the correct power supply for that GPIO port?	☐	
4	If the Tamper pins are to be used, have they been properly connected?	☐	
5	Do Tamper pins 0, 1, and 2 have a note to bury them between ground planes? Alternatively (if this cannot be done) 3 pF bypass capacitors can be added to each pin.	☐	
6	Has the VBAT_WAKEUP_b pin (P5_2) been properly handled for the application? (This pin drives low on reset.)	☐	
7	Does the design ensure that GPIO will not be driven before the application of power to the supplies?	☐	

Table 8. ADC/DAC checks

S. No.	Check description	Check done (Y/ N)	Remarks / Actions
1	Does the ADC characteristics fit the application?	☐	
2	Has the VREFH voltage been chosen according to the desired ADC range?	☐	
3	Does the input impedance to the ADC channels fall under the maximum ADC impedance?	☐	

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Table 8. ADC/DAC checks*...continued*

S. No.	Check description	Check done (Y/ N)	Remarks / Actions
4	Has the VREFH voltage been chosen according to the desired DAC range?	☐	
5	Have the load capacitors for the DAC been included?	☐	

Table 9. VREFOUT check

S. No.	Check description	Check done (Y/ N)	Remarks / Actions
1	If the VREF_OUT functionality is used for either an internal or external reference, has a 100 nF capacitor to GND been connected to VREF_OUT?	☐	

Table 10. Programming and debug checks

S. No.	Check description	Check done (Y/ N)	Remarks / Actions
1	Has the debug connector been connected to the correct I/O on the microcontroller?	☐	
2	Has a connector been added that allows the microcontroller to be put into programming mode?	☐	
3	Is RESET_b properly connected to the debug connector?	☐	

Table 11. External memory related checks

S. No.	Check description	Check done (Y/ N)	Remarks / Actions
1	Is the external memory used a supported memory type?	☐	
2	If used for boot, is it connected to FlexSPI A?	☐	
3	If using DDR mode, is DQS connected to the memory device's DQS pin and only this pin?	☐	
4	Has the flash memory symbol pinning been checked?	☐	

Table 12. Reset and ISP pin checks

S. No.	Check description	Check done (Y/ N)	Remarks / Actions
1	Has the ISP mode pin been configured appropriately?	☐	
2	Have capacitors been placed on the reset pin?	☐	
3	If using ISP mode, are the bootloader pins correctly connected for the chosen interface(s)?	☐	

Table 13. USB checks

S. No.	Check description	Check done (Y/ N)	Remarks / Actions
USB FS			
1	Has the VBUS pin been properly connected through a voltage divider circuit?	☐	
2	Have the inline 33 ohm resistors been placed with the CAD note that they should be placed as close as possible to the USB0_DM and USB0_DP pins?	☐	

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Table 13. USB checks*...continued*

S. No.	Check description	Check done (Y/ N)	Remarks / Actions
3	Has the USB circuit been properly decoupled (if used)?	☐	
4	Have transient protection diodes been placed on USB data lines and supply lines (if using USB)?	☐	
5	Has the USB shield (casing) been properly grounded (if using USB)?	☐	
USB HS			
1	Has the VBUS pin been directly connected to the USB VBUS?	☐	
2	Have the inline resistors been from USB0_DM and USB0_DP pins?	☐	
3	Has the ID pin been appropriately connected for OTG applications?	☐	

12.2 Layout checklists

This section provides the checklists that can be used for layout review.

Table 14. General layout checks

Serial No	Check Description	Check done (Y/N)	Remarks/ Action
1	Has the number of layers been clearly discussed?	☐	
2	Has the layout been checked versus NXP reference board (latest revision)?	☐	
3	Has the correct PCB material been specified?	☐	
4	Have the correct PCB thicknesses been specified?	☐	

Table 15. High frequency crystal reference oscillator

Serial No	Check Description	Check done (Y/N)	Remarks/ Action
1	Has the XTAL been placed close to the IC?	☐	
2	Have GND vias been placed around the XTAL?	☐	

Table 16. Power supply checks

Serial No	Check Description	Check done (Y/N)	Remarks/ Action
1	Have all the decoupling capacitors been placed as close as possible to the power pins and voltage regulators outputs?	☐	
2	Have multiple vias been added to interconnect power lines when power is routed on several layers?	☐	
3	For low supply voltage application – close to 1.71 V, ensure that the supply track is well dimensioned to avoid IR drop and therefore falsely trigger the brownout. (IR drop is the voltage drop due to the supply current "I" flowing into the supply track resistance "R").	☐	
4	Have the VDD lines been isolated from potential interferences on all layers?	☐	
5	Have the crystal references been placed on the same side of the board as the MCU?	☐	

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Table 16. Power supply checks*...continued*

Serial No	Check Description	Check done (Y/N)	Remarks/ Action
6	If the DCDC regulator is used, has the inductor been placed on the same side of the board as the MCU?	☐	
7	If the DCDC regulator is used, has the recommended DCDC layout been followed?	☐	

Table 17. EMC and miscellaneous checks

Serial No	Check Description	Check done (Y/N)	Remarks/ Action
1	If more than 2 layers are used, does the second layer below the top/component side act as a continuous ground plane (GND reference plane)?	☐	
2	Have traces been avoided below noisy or sensitive components?	☐	
3	Do traces cut through power or ground planes unnecessarily?	☐	
4	Has a GND area or trace been added to isolate one signal from another one, where possible? (The ground area or trace must be properly connected to the reference GND plane with vias; otherwise it is better to remove it.)	☐	
5	Have all the connections between a component and GND be doubled with a via to the GND plane?	☐	
6	Have the small GND fills and isolated fingers that cannot be connected to the reference GND plane with a via been removed?	☐	
7	Have silk screens been added with relevant information? (components ref, logo, board name, and so on)	☐	
8	Are all silkscreen texts readable when the board is populated?	☐	
9	Is the footprint strictly similar to the NXP reference?	☐	
10	Are copper wires or areas clearance to the PCB edge respected according to the PCB manufacturer rules?	☐	
11	Have the soldering/non-soldering areas been respected? Is the solder resist layer checked in the empty area?	☐	
12	If using the DCDC regulator, have the charge / discharge loops been minimized as per the Hardware Design Guidelines?	☐	

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13 References

Table 18 lists documents and resources that can be referred to for more information. Some of the documents listed below are available only under a non-disclosure agreement (NDA). To request access to these documents, contact your local field applications engineer (FAE) or sales representative.

Table 18. Related documentation

Document	Link/how to access
• *MCX N series Data Sheet* (Document ID MCXNP184M150F70) • MCXN23x Data Sheet (Document ID MCXN23x)	MCXNP184M150F70 MCXN23x
• *MCX N series Reference Manual* (Document ID MCXNP184M150F70 RM • MCXN23XRM (Document ID MCXN23XRM)	MCXNP184M150F70RM MCXN23XRM
• *MCX N Chip Errata* (Document ID MCXN_1P02G • *MCX N23x Chip Errata* (Document ID MCXN23x_0P21K)	MCXN_1P02G MCXN23x_0P21K
• *MCX N Series Security Reference Manual* (Document ID MCXNP184 M150F70SRM) • MCX N23x Security Reference Manual (Document ID MCXN23xSRM)	MCXNP184M150F70SRM MCXN23XSRM

13.1 More resources

MCX N94, N54, N53, N52, and N24

- For an overview of the MCX N94, N54, N53, N52, and N24 MCUs, refer: MCX N Product Overview
- Additional documents are available on the URL: MCX N Documentation
- Design resources are available on the URL: MCX N Design resources
- Design checklist: MCX_N_Series_Design_In_checklist_v4.xlsx

MCX N23

- For an overview of the MCX N23 devices, refer: MCX N23x
- Additional documents are available on the URL: MCX-N23X Documentation
- Design resources are available on the URL: MCX N23x Design resources
- Design checklist: MCX_N_Series_Design_In_checklist_v4.xlsx

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14 Acronyms and abbreviations

Table 19 defines the acronyms and abbreviations used in this document.

Table 19. Acronyms and abbreviations

Acronym	Definition
ADC	Analog-to-digital converter
BGA	Ball grid array
BoM	Bill of materials
DAC	Digital-to-analog converter
DCDC	Direct current-to-direct current
DDR	Double data rate
DMA	Direct memory access
DNP	Do not populate
EMC	Electromagnetic compatibility
EMI	Electromagnetic interference
eMMC	Embedded multimedia card
ESD	Electrostatic discharge
GPIO	General-purpose input/output
HDQFP	High-density quad flat package
HLQFP	Thermal enhanced low profile quad flat package
HS USB	High-speed USB
I/O	Input/output
I2C	Inter-Integrated circuit
IEEE	Institute of Electrical and Electronics Engineers
ISP	In-system programming
JTAG	Joint test access group
LAN	Local area network
LDO	Low drop-out
MAC	Media access controller
MCU	Microcontroller unit
MII	Media-independent interface
PCB	Printed circuit board
PHY	Physical interface of the OSI model
PLL	Phase-locked loop
QSPI	Quadruple serial peripheral interface
RC circuit	Resistor–capacitor circuit
RMII	Reduced media independent interface
SPI	Serial peripheral interface

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Table 19. Acronyms and abbreviations...*continued*

Acronym	Definition
TSI	Touch system interface
TVS	Transient voltage suppression
USB	Universal serial bus
VFBGA	Very thin profile fine pitch ball grid array
VIP	Via-in-pad
VREF	Voltage reference

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15 Revision history

Table 20 summarizes the revisions to this document.

Table 20. Revision history

Document ID	Release date	Description
UG10092 v. 6.1	30 June 2026	• Corrected the Figure 4. • Updated VDD to VDD_SYS in the Section 5.2 "Mixed supply configuration" See List item 1 under Power sequencing rules.
UG10092 v. 6.0	3 June 2026	Updated Figure 12 (HDQFP172 LDO configuration).
UG10092 v. 5.0	30 January 2026	• Added and updated the document to include guidelines for all MCX N devices (N94, N54, N53, N52, N24, N23). • Updated Section 12 "Design checklists".
UG10092 v. 4.1	7 April 2025	Updated Figure 10 and Section 3.2.4 "HLQFP routing".
UG10092 v. 4.0	17 February 2025	• Added Section 3.3 "HDQFP172 package". • Added link in Section 12 "Design checklists". • Updated to include 172-pin HDQFP. • Updated remarks for VDD_SYS in Table 1. • Updated Section 12.1 "Schematic checklist"Section 12.2 "Layout checklists". • Added Design checklists link in Section 13 "References". • Added Figure 18.
UG10092 v. 3.0	22 October 2024	• Updates in Section 5.1 "Single supply configuration". • Other minor updates.
UG10092 v. 2.0	1 August 2024	• Added the section Section 12 "Design checklists". • Updated figures in Section 2 "Architecture overview". • Other updates.
UG10092 v.1.0	19 January 2024	Initial public release.

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